Partial Differential Equations

Course Code: MSMAT03DSC12

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0 Preliminaries

This section collects the notation and the facts from analysis that are used throughout the notes. Later sections refer back to the numbered results here. Results stated "without proof" are standard facts from calculus and measure theory.

0.1 Notation

Definition 0.1 (Balls, boundaries, distances)**.** Let $x \in \mathbb{R}^n$, $r > 0$ and $U \subset \mathbb{R}^n$.

1. $|x| = (x_1^2 + \cdots + x_n^2)^{1/2}$ and $x \cdot y = \sum_{i=1}^n x_i y_i$.
2. $B(x,r) = \{y \in \mathbb{R}^n : |y - x| \le r\}$ is the *closed* ball with center x and radius r, and $\partial B(x,r) = \{y : |y - x| = r\}$ is the sphere. The open ball $\{y : |y - x| < r\}$ is denoted $B^\circ(x,r)$. (For integrals it makes no difference whether we integrate over $B(x,r)$ or $B^\circ(x,r)$, since the sphere has volume zero.)
3. $\bar{U}$ is the closure of U, and $\partial U = \bar{U} \setminus U^\circ$ is its boundary. If U is open, $\partial U = \bar{U} \setminus U$.
4. $\operatorname{dist}(x,E) = \inf\{|x - y| : y \in E\}$ for $E \neq \emptyset$, and $\operatorname{dist}(x,\emptyset) = +\infty$. Also $\operatorname{dist}(V,E) = \inf\{|x - y| : x \in V, y \in E\}$.
5. We write $V \subset\subset U$ ("V is compactly contained in U") if $V \subset \bar{V} \subset U$ and $\bar{V}$ is compact.
6. If U is open and $\varepsilon > 0$, we write $U_\varepsilon := \{x \in U : \operatorname{dist}(x, \partial U) > \varepsilon\}$. If $U = \mathbb{R}^n$ then $\partial U = \emptyset$ and $U_\varepsilon = \mathbb{R}^n$.

The following lemma is used every time we need "a ball which lies completely inside the domain".

Lemma 0.2 (Balls inside an open set)**.** *Let $U \subset \mathbb{R}^n$ be open and $x \in U$.*

(a) If $0 < r < \operatorname{dist}(x, \partial U)$, then the closed ball $B(x,r) \subset U$.

(b) Conversely, if the closed ball $B(x,r) \subset U$ and $\partial U \neq \emptyset$, then $r < \operatorname{dist}(x, \partial U)$.

(c) The set U_ε is open, and the function $x \mapsto \operatorname{dist}(x, \partial U)$ satisfies $|\operatorname{dist}(x, \partial U) - \operatorname{dist}(x', \partial U)| \le |x - x'|$.

Proof. (a) Suppose some y with $|y - x| \le r$ is not in U. Move along the segment $p(t) = x + t(y - x)$, $0 \le t \le 1$, and let $t^* = \sup\{t \in [0,1] : p(s) \in U \text{ for all } s \in [0,t]\}$. The point $p(t^*)$ is a limit of points $p(s) \in U$, so $p(t^*) \in \bar{U}$. If $p(t^*)$ were in U, then (as U is open) a small piece of the segment beyond t^* would also lie in U, which contradicts the definition of t^* unless $t^* = 1$; but $t^* = 1$ would give $p(1) = y \in U$, which is false. Hence $p(t^*) \in \bar{U} \setminus U = \partial U$. But $|p(t^*) - x| = t^*|y - x| \le r < \operatorname{dist}(x, \partial U)$, a contradiction.

(b) Every point of ∂U lies outside U (as U is open), hence outside $B(x,r)$, so $|x - z| > r$ for all $z \in \partial U$. The function $z \mapsto |x - z|$ attains its infimum on the closed set ∂U (intersect ∂U with a large closed ball to get a compact set), so $\operatorname{dist}(x, \partial U) > r$.

(c) For $z \in \partial U$, $|x - z| \le |x - x'| + |x' - z|$. Taking the infimum over z gives $\operatorname{dist}(x, \partial U) \le |x - x'| + \operatorname{dist}(x', \partial U)$, and by symmetry the stated inequality. So $x \mapsto \operatorname{dist}(x, \partial U)$ is continuous and $U_\varepsilon = U \cap \{x : \operatorname{dist}(x, \partial U) > \varepsilon\}$ is the intersection of two open sets. □

Definition 0.3 (Derivatives and function spaces)**.** Let $U \subset \mathbb{R}^n$ be open and $u : U \to \mathbb{R}$.

1. $u_{x_i} = \frac{\partial u}{\partial x_i}$, $u_{x_i x_j} = \frac{\partial^2 u}{\partial x_i \partial x_j}$, etc. $e_i = (0, \dots, 1, \dots, 0)$ is the ith coordinate vector.
2. A *multiindex* is $\alpha = (\alpha_1, \dots, \alpha_n)$ with each α_i a nonnegative integer; its order is $|\alpha| = \alpha_1 + \cdots + \alpha_n$, and
$$D^\alpha u = \frac{\partial^{|\alpha|} u}{\partial x_1^{\alpha_1} \cdots \partial x_n^{\alpha_n}}.$$
$D^k u = \{D^\alpha u : |\alpha| = k\}$ and $|D^k u| = \big(\sum_{|\alpha|=k} |D^\alpha u|^2\big)^{1/2}$. In particular $Du = (u_{x_1}, \dots, u_{x_n})$ is the gradient, $D^2 u$ is the Hessian matrix, and $\Delta u = \sum_{i=1}^n u_{x_i x_i}$ is the Laplacian.
3. $C^k(U)$ is the set of functions whose partial derivatives of order $\le k$ exist and are continuous in U; $C^\infty(U) = \bigcap_k C^k(U)$. $C^k(\bar{U})$ consists of those $u \in C^k(U)$ for which every $D^\alpha u$, $|\alpha| \le k$, extends continuously to $\bar{U}$. For $u \in C^2(U)$ mixed partials are equal: $u_{x_i x_j} = u_{x_j x_i}$.
4. The *support* of u is $\operatorname{spt} u = \overline{\{x : u(x) \neq 0\}}$, the closure of the set where u is nonzero. u has *compact support* if $\operatorname{spt} u$ is compact, i.e. (in $\mathbb{R}^n$) if u vanishes outside some ball. $C_c^k(U)$ is the set of $u \in C^k(U)$ with compact support contained in U.
5. For a set E: $\|u\|_{L^\infty(E)} = \sup_E |u|$ and $\|u\|_{L^1(E)} = \int_E |u|\, dx$.

Definition 0.4 (Averages)**.**

$$\fint_{B(x,r)} f\, dy = \frac{1}{\alpha(n) r^n} \int_{B(x,r)} f\, dy = \text{average of } f \text{ over the ball } B(x,r),$$

$$\fint_{\partial B(x,r)} f\, dS = \frac{1}{n\alpha(n) r^{n-1}} \int_{\partial B(x,r)} f\, dS = \text{average of } f \text{ over the sphere } \partial B(x,r).$$

Here $\int_\Sigma f\, dS$ denotes the integral of f over the surface Σ with respect to $(n-1)$-dimensional surface measure. Since $\alpha(n) r^n$ and $n\alpha(n) r^{n-1}$ are the volume of the ball and the area of the sphere (Fact 0.5), the average of the constant function 1 is 1. Consequently, for any constant c, $\fint c = c$ and $\fint (f - c) = \fint f - c$.

0.2 Volumes and Surface Areas of Balls

Fact 0.5 (without proof)**.** Let Γ denote the Gamma function, $\Gamma(s) = \int_0^\infty t^{s-1} e^{-t}\, dt$.

1. The volume of the unit ball $B(0,1)$ in $\mathbb{R}^n$ is
$$\alpha(n) = \frac{\pi^{n/2}}{\Gamma(\frac{n}{2} + 1)}.$$
2. The surface area of the unit sphere $\partial B(0,1)$ in $\mathbb{R}^n$ is $n\alpha(n)$.
3. The volume of a ball of radius r (with any center) is $|B(x,r)| = \alpha(n) r^n = r^n\, |B(0,1)|$.
4. The surface area of a sphere of radius r is $|\partial B(x,r)| = n\alpha(n) r^{n-1} = r^{n-1} |\partial B(0,1)|$.
5. Volume and area are related by
$$|\partial B(x,r)| = \frac{n}{r} |B(x,r)|, \qquad \frac{d}{dr} |B(x,r)| = \frac{d}{dr}\big(\alpha(n) r^n\big) = n\alpha(n) r^{n-1} = |\partial B(x,r)|.$$
For $r = 1$: (area of unit sphere) $= n\times$ (volume of unit ball).

Small dimensions: $\alpha(1) = 2$, $\alpha(2) = \pi$, $\alpha(3) = \frac{4\pi}{3}$; the corresponding "surface areas" $n\alpha(n)$ are 2 (two points), 2π (circumference of the unit circle) and 4π. The scaling rules (3), (4) say: stretching by the factor r multiplies n-dimensional volume by r^n and $(n-1)$-dimensional area by r^{n-1}.

0.3 Polar Coordinates

Theorem 0.6 (Polar coordinates)**.** *(i) Let $f : \mathbb{R}^n \to \mathbb{R}$ be continuous and summable. Then*

$$\int_{\mathbb{R}^n} f\,dx = \int_0^\infty \left(\int_{\partial B(x_0,r)} f\,dS \right) dr$$

for each point $x_0 \in \mathbb{R}^n$.

(ii) In particular

$$\frac{d}{dr}\left(\int_{B(x_0,r)} f\,dx \right) = \int_{\partial B(x_0,r)} f\,dS$$

for each $r > 0$.

The ball is the union of the spheres $\partial B(x_0, r)$, so we integrate first over each sphere and then over the radius. We use the following consequences (without proof).

Corollary 0.7 (Polar coordinates on balls; scaling)**.** *Let $x_0 \in \mathbb{R}^n$ and $R > 0$.*

(a) Balls with any center. *If f is continuous on $B(x_0, R)$, then*

$$\int_{B(x_0,R)} f\,dx = \int_0^R \left(\int_{\partial B(x_0,r)} f\,dS \right) dr.$$

(b) Reduction to the unit ball and the unit sphere. *Substituting $y = x_0 + rz$ (so z runs over $B(0,1)$ or $\partial B(0,1)$):*

$$\int_{B(x_0,r)} f(y)\,dy = r^n \int_{B(0,1)} f(x_0 + rz)\,dz,$$
$$\int_{\partial B(x_0,r)} f(y)\,dS(y) = r^{n-1} \int_{\partial B(0,1)} f(x_0 + rz)\,dS(z).$$

Dividing by $\alpha(n)r^n$, resp. $n\alpha(n)r^{n-1}$, the factors r^n, r^{n-1} cancel and

$$\fint_{B(x_0,r)} f(y)\,dy = \fint_{B(0,1)} f(x_0+rz)\,dz, \qquad \fint_{\partial B(x_0,r)} f(y)\,dS(y) = \fint_{\partial B(0,1)} f(x_0+rz)\,dS(z).$$

(c) Radial functions. *If $f(y) = g(|y - x_0|)$ depends only on the distance to x_0, then f is constant ($= g(r)$) on each sphere $\partial B(x_0, r)$, so*

$$\int_{\partial B(x_0,r)} f\,dS = n\alpha(n) r^{n-1} g(r), \qquad \int_{B(x_0,R)} f\,dy = n\alpha(n) \int_0^R g(r)\, r^{n-1}\,dr.$$

The second formula remains true (both sides possibly $+\infty$) if $g \geq 0$ is only continuous on $(0, R]$: apply (a) on the annulus $\delta \leq |y - x_0| \leq R$ and let $\delta \to 0$, using the Monotone Convergence Theorem (Fact 0.16).

0.4 Integrable and Locally Integrable Functions

Definition 0.8. Let $E \subset \mathbb{R}^n$ be measurable and $U \subset \mathbb{R}^n$ open.

1. f is *integrable* (or *summable*) on E, written $f \in L^1(E)$, if $\int_E |f|\,dx < \infty$. For $1 \leq p < \infty$, $f \in L^p(E)$ if $\|f\|_{L^p(E)} = \left(\int_E |f|^p dx \right)^{1/p} < \infty$.

2. f is *locally integrable* on U, written $f \in L^1_{\text{loc}}(U)$, if $f \in L^1(K)$ for every compact set $K \subset U$ (equivalently, $f \in L^1(V)$ for each $V \subset\subset U$). Similarly $f \in L^p_{\text{loc}}(U)$ if $f \in L^p(K)$ for every compact $K \subset U$.

Integrability is a condition on the whole set; local integrability only asks that f is not "too singular" anywhere inside U (it may still grow at infinity or near ∂U).

Example 0.9. Recall that a compact set K is bounded, so it has finite volume $|K|$.

1. *Bounded functions are locally integrable.* If $|f| \le M$ (and f is measurable), then $\int_K |f| \le M|K| < \infty$.

2. *Constant functions are locally integrable*, by (1). The constant 1 on $\mathbb{R}^n$ is *not* integrable: $\int_{\mathbb{R}^n} 1\,dx = \infty$.

3. *Continuous functions are locally integrable.* A continuous f on U is bounded on each compact $K \subset U$ (it attains its maximum $\max_K |f|$), so $\int_K |f| \le (\max_K |f|)\,|K|$. Examples which are locally integrable but not integrable on $\mathbb{R}$: $f(x) = x$, polynomials, e^x, $\sin x$ (note $\int_{\mathbb{R}} |\sin x|\,dx = \infty$).

4. *Locally integrable depends on the domain.* $f(x) = 1/x$ is locally integrable on $U = (0,\infty)$ (on each compact $K \subset (0,\infty)$ it is continuous), but $f(x) = 1/|x|$ is *not* locally integrable on $\mathbb{R}$, since $\int_{-1}^1 \frac{dx}{|x|} = 2\int_0^1 \frac{dx}{x} = \infty$.

5. *Powers of $|x|$.* By Lemma 0.10 below, $|x|^{-m}$ is locally integrable on $\mathbb{R}^n$ exactly when $m < n$, and it is never integrable on $\mathbb{R}^n$. For instance $1/|x|$ in $\mathbb{R}^3$ is locally integrable but not integrable, and $1/|x|^n$ in $\mathbb{R}^n$ is not locally integrable.

6. Continuous functions with compact support are integrable: $\int_{\mathbb{R}^n} |f| = \int_{\operatorname{spt} f} |f| < \infty$ by (3).

Lemma 0.10 (Integrability of $|x|^{-m}$)**.** *Let $m \in \mathbb{R}$ and consider $f(x) = |x|^{-m}$ for $x \in \mathbb{R}^n$, $x \neq 0$. Then*

(i) $\displaystyle\int_{B(0,1)} \frac{dx}{|x|^m} < \infty$ *if and only if $m < n$; in that case it equals* $\dfrac{n\alpha(n)}{n-m}$;

(ii) $\displaystyle\int_{\mathbb{R}^n \setminus B(0,1)} \frac{dx}{|x|^m} < \infty$ *if and only if $m > n$; in that case it equals* $\dfrac{n\alpha(n)}{m-n}$;

(iii) $|x|^{-m}$ *is never integrable on all of $\mathbb{R}^n$;*

(iv) $|x|^{-m}$ *is locally integrable on $\mathbb{R}^n$ if and only if $m < n$. The same holds for $|x - x_0|^{-m}$, for any fixed x_0.*

Proof. The function is radial with $g(r) = r^{-m}$, and positive, so Corollary 0.7(c) applies. Put $p = n - 1 - m$, so that $g(r)r^{n-1} = r^p$.

(i) $\int_{B(0,1)} |x|^{-m} dx = n\alpha(n)\int_0^1 r^p\,dr = n\alpha(n)\lim_{\delta\to 0^+}\int_\delta^1 r^p\,dr$. We compute:

- If $p \neq -1$: $\int_\delta^1 r^p\,dr = \frac{1-\delta^{p+1}}{p+1}$. If $p > -1$ then $\delta^{p+1} \to 0$ and the limit is $\frac{1}{p+1}$. If $p < -1$ then $\delta^{p+1} = 1/\delta^{|p+1|} \to \infty$, and the limit is $+\infty$.
- If $p = -1$: $\int_\delta^1 r^{-1} dr = -\log\delta \to +\infty$.

So the integral is finite iff $p > -1$, i.e. $n-1-m > -1$, i.e. $m < n$, and then it equals $\frac{n\alpha(n)}{p+1} = \frac{n\alpha(n)}{n-m}$.

(ii) Similarly, $\int_{|x|\geq 1} |x|^{-m} dx = n\alpha(n) \lim_{R\to\infty} \int_1^R r^p \, dr$. If $p \neq -1$, $\int_1^R r^p dr = \frac{R^{p+1}-1}{p+1}$, which converges (to $\frac{-1}{p+1}$) iff $p+1 < 0$; if $p = -1$ it equals $\log R \to \infty$. So the integral is finite iff $p < -1$, i.e. $m > n$, and then it equals $\frac{n\alpha(n)}{-(p+1)} = \frac{n\alpha(n)}{m-n}$.

(iii) Integrability on $\mathbb{R}^n$ would require both (i) and (ii), i.e. $m < n$ and $m > n$, which is impossible.

(iv) If $m < n$ and K is compact, choose $R \geq 1$ with $K \subset B(0,R)$. Then, as in (i), $\int_K |x|^{-m} \leq n\alpha(n) \int_0^R r^p \, dr = \frac{n\alpha(n)R^{n-m}}{n-m} < \infty$. If $m \geq n$, then $K = B(0,1)$ already gives an infinite integral by (i). The statement for $|x-x_0|^{-m}$ follows by the translation $x \mapsto x - x_0$, which does not change integrals. □

0.5 Limits, Derivatives and Integrals

Many proofs need to exchange a limit (or a derivative) with an integral. This is *not* automatic; the following lemmas say when it is allowed. Whenever we exchange limits and integrals later, we refer to one of these.

Lemma 0.11 (A nonnegative continuous function with zero integral vanishes)**.** *Let g be continuous on the closed ball $B(x,r)$, $r > 0$, with $g \geq 0$ and $\int_{B(x,r)} g \, dy = 0$. Then $g \equiv 0$ on $B(x,r)$. Equivalently: if $g \geq 0$ is continuous on $B(x,r)$ and $g(y_0) > 0$ for some y_0, then $\int_{B(x,r)} g \, dy > 0$.*

Proof. Suppose $g(y_0) > 0$ for some y_0 with $|y_0 - x| < r$. By continuity there is $\delta > 0$ with $B(y_0,\delta) \subset B(x,r)$ and $g > g(y_0)/2$ on $B(y_0,\delta)$. Since $g \geq 0$,

$$\int_{B(x,r)} g \, dy \geq \int_{B(y_0,\delta)} g \, dy \geq \frac{g(y_0)}{2} \, \alpha(n)\delta^n > 0,$$

a contradiction. So $g = 0$ on the open ball $B^\circ(x,r)$. Every point of the sphere $\partial B(x,r)$ is a limit of points of the open ball, so by continuity $g = 0$ there too. □

Lemma 0.12 (Uniform convergence and integrals)**.** *Let E be a measurable set (or a surface such as a sphere, with surface measure), let $w \in L^1(E)$, and let g_h, g be bounded measurable functions on E with $g_h \to g$* uniformly *on E as $h \to 0$, that is, $\sup_E |g_h - g| \to 0$. Then*

$$\int_E g_h \, w \;\longrightarrow\; \int_E g \, w \qquad (h \to 0).$$

In particular, if E has finite measure, we may take $w \equiv 1$: $\int_E g_h \to \int_E g$.

Proof. $\left| \int_E g_h w - \int_E g w \right| \leq \int_E |g_h - g| \, |w| \leq \Big(\sup_E |g_h - g| \Big) \int_E |w| \to 0.$ □

Lemma 0.13 (Difference quotients converge uniformly)**.** *(a) If $g : \mathbb{R}^n \to \mathbb{R}$ is uniformly continuous, then $\sup_{z\in\mathbb{R}^n} |g(z+a) - g(z)| \to 0$ as $a \to 0$.*

(b) If $g \in C^1(\mathbb{R}^n)$ and g_{x_i} is uniformly continuous on $\mathbb{R}^n$, then

$$\sup_{z\in\mathbb{R}^n} \left| \frac{g(z+he_i) - g(z)}{h} - g_{x_i}(z) \right| \to 0 \qquad (h \to 0).$$

Both hypotheses hold if g (resp. g_{x_i}) is continuous with compact support, because a continuous function with compact support is uniformly continuous on $\mathbb{R}^n$.

Proof. (a) is the definition of uniform continuity: given $\eta > 0$ there is $\delta > 0$ such that $|g(z + a) - g(z)| < \eta$ for all z whenever $|a| < \delta$.

(b) Fix z and apply the one-variable mean value theorem to $t \mapsto g(z + te_i)$ on the interval between 0 and h: there is $\theta = \theta(z, h) \in (0, 1)$ with

$$\frac{g(z + he_i) - g(z)}{h} = g_{x_i}(z + \theta he_i).$$

Hence the quantity in (b) equals $|g_{x_i}(z+\theta he_i) - g_{x_i}(z)|$. The two points differ by θhe_i, of length $< |h|$, so by uniform continuity of g_{x_i} this is $< \eta$ for *all* z once $|h| < \delta$. □

Lemma 0.14 (Differentiation under the integral sign over a compact set)**.** *Let K be a compact subset of $\mathbb{R}^n$ (with volume measure) or a sphere $\partial B(0, 1)$ (with surface measure), and let $I \subset \mathbb{R}$ be an open interval. Let $F : I \times K \to \mathbb{R}$ be continuous, and suppose $\frac{\partial F}{\partial t}(t, z)$ exists and is continuous on $I \times K$. Then $\Psi(t) = \int_K F(t, z)\, dz$ is differentiable on I, and*

$$\Psi'(t) = \int_K \frac{\partial F}{\partial t}(t, z)\, dz.$$

Proof. Fix $t_0 \in I$ and choose $[a, b] \subset I$ with $a < t_0 < b$. The set $[a, b] \times K$ is compact, so $\partial_t F$ is uniformly continuous there. For small $h \neq 0$ the mean value theorem in t gives, for each z,

$$\frac{F(t_0 + h, z) - F(t_0, z)}{h} = \frac{\partial F}{\partial t}(t_0 + \theta h, z), \qquad \theta = \theta(z, h) \in (0, 1).$$

By uniform continuity, $\left|\partial_t F(t_0 + \theta h, z) - \partial_t F(t_0, z)\right| < \eta$ for all $z \in K$ when $|h|$ is small. Thus the difference quotients converge to $\partial_t F(t_0, \cdot)$ uniformly on K, and since K has finite measure Lemma 0.12 (with $w \equiv 1$) gives

$$\frac{\Psi(t_0 + h) - \Psi(t_0)}{h} = \int_K \frac{F(t_0 + h, z) - F(t_0, z)}{h}\, dz \to \int_K \frac{\partial F}{\partial t}(t_0, z)\, dz. \qquad \square$$

Lemma 0.15 (Differentiating a convolution)**.** *Let $U \subset \mathbb{R}^n$ be open (possibly $U = \mathbb{R}^n$), $\rho > 0$, $f \in L^1_{\mathrm{loc}}(U)$, and let $g \in C^1(\mathbb{R}^n)$ with $g(z) = 0$ for $|z| > \rho$. For $x \in U_\rho$ define*

$$F(x) = \int_U g(x - y) f(y)\, dy.$$

Then:

(a) F is well defined and continuous on U_ρ;

(b) $F \in C^1(U_\rho)$ and $\dfrac{\partial F}{\partial x_i}(x) = \displaystyle\int_U \frac{\partial g}{\partial x_i}(x - y) f(y)\, dy$;

(c) if $g \in C^k(\mathbb{R}^n)$, then $F \in C^k(U_\rho)$ and $D^\alpha F(x) = \int_U D^\alpha g(x - y) f(y)\, dy$ for $|\alpha| \leq k$. If $g \in C^\infty$, then $F \in C^\infty(U_\rho)$.

Proof. Where the integrand lives. Fix $x \in U_\rho$ and put $d = \operatorname{dist}(x, \partial U) - \rho > 0$ (if $U = \mathbb{R}^n$, put $d = 1$). Let $K = B(x, \rho + d/2)$. Since $\rho + d/2 < \operatorname{dist}(x, \partial U)$, Lemma 0.2(a) gives $K \subset U$; K is compact, so $\int_K |f| < \infty$. Now let $|a| \leq d/2$. Then $\operatorname{dist}(x + a, \partial U) \geq \operatorname{dist}(x, \partial U) - |a| > \rho$ (Lemma 0.2(c)), so $x + a \in U_\rho$; and $g(x + a - y) \neq 0$ forces $|x + a - y| \leq \rho$, hence $|y - x| \leq \rho + |a| \leq \rho + d/2$, i.e. $y \in K$. Therefore

$$F(x + a) = \int_K g(x + a - y) f(y)\, dy \qquad (|a| \leq d/2),$$

and the same holds with g replaced by any derivative of g (which also vanishes for $|z| > \rho$).

(a) g is continuous and vanishes outside $B(0,\rho)$, so it is bounded, $|g| \le M$, and $\int_K |g(x-y)f(y)|\,dy \le M\int_K |f| < \infty$. Also g is uniformly continuous (Lemma 0.13), and with $z = x - y$,

$$\begin{aligned}|F(x+a) - F(x)| &\le \int_K |g(x-y+a) - g(x-y)|\,|f(y)|\,dy \\ &\le \Big(\sup_z |g(z+a) - g(z)|\Big)\int_K |f| \to 0 \quad (a \to 0).\end{aligned}$$

(b) For $0 < |h| \le d/2$, put $M_h(z) = \frac{g(z+he_i)-g(z)}{h} - g_{x_i}(z)$. Then

$$\begin{aligned}\left|\frac{F(x+he_i)-F(x)}{h} - \int_K g_{x_i}(x-y)f(y)\,dy\right| &= \left|\int_K M_h(x-y)f(y)\,dy\right| \\ &\le \Big(\sup_z |M_h(z)|\Big)\int_K |f| \to 0\end{aligned}$$

by Lemma 0.13(b) (g_{x_i} is continuous and vanishes for $|z| > \rho$, hence uniformly continuous). So $F_{x_i}(x)$ exists and equals $\int_U g_{x_i}(x-y)f(y)\,dy$. Applying (a) to g_{x_i} in place of g shows that F_{x_i} is continuous.

(c) If $g \in C^k$, each g_{x_i} is in C^{k-1} and vanishes for $|z| > \rho$; apply (b) to g_{x_i}, and repeat. By induction on $|\alpha|$ we get the formula for $D^\alpha F$ and its continuity. □

Fact 0.16 (Facts from measure theory, without proof)**.** 1. *Monotone Convergence Theorem.* If $0 \le f_1 \le f_2 \le \cdots$ and $f_k \to f$ a.e., then $\int f_k \to \int f$.

2. *Dominated Convergence Theorem.* If $f_k \to f$ a.e. and $|f_k| \le g$ with g integrable, then $\int f_k \to \int f$.

3. *Fubini–Tonelli.* If $F(x,y) \ge 0$ is measurable, then $\iint F(x,y)\,dy\,dx = \iint F(x,y)\,dx\,dy$ (the order of integration may be exchanged).

4. *Hölder's inequality.* If $1 < p < \infty$ and $\frac{1}{p} + \frac{1}{q} = 1$, then $\int |fg| \le \big(\int |f|^p\big)^{1/p}\big(\int |g|^q\big)^{1/q}$.

5. *Density.* If W is open and $1 \le p < \infty$, then $C_c(W)$ is dense in $L^p(W)$: for each $f \in L^p(W)$ and $\delta > 0$ there is a continuous g with compact support in W and $\|f - g\|_{L^p(W)} < \delta$.

Theorem 0.17 (Lebesgue's Differentiation Theorem)**.** *Let $f : \mathbb{R}^n \to \mathbb{R}$ be locally summable.*

(i) Then for a.e. point $x_0 \in \mathbb{R}^n$,

$$⨍_{B(x_0,r)} f\,dx \to f(x_0) \quad \text{as } r \to 0.$$

(ii) In fact, for a.e. point $x_0 \in \mathbb{R}^n$,

$$⨍_{B(x_0,r)} |f(x) - f(x_0)|\,dx \to 0 \quad \text{as } r \to 0. \tag{1}$$

A point x_0 at which (1) holds is called a *Lebesgue point* of f.

Remark 0.18. (1) More generally, if $f \in L^p_{\mathrm{loc}}(\mathbb{R}^n)$ for some $1 \le p < \infty$, then for a.e. point $x_0 \in \mathbb{R}^n$ we have $⨍_{B(x_0,r)} |f(x) - f(x_0)|^p\,dx \to 0$ as $r \to 0$.

(2) *The theorem is local*, so it also holds for $f \in L^1_{\mathrm{loc}}(U)$, U open: for a.e. $x_0 \in U$, (1) holds. Indeed, let $B_1, B_2, \ldots$ be the countably many open balls with rational centers and rational radii whose closures lie in U; they cover U. Apply the theorem to $f_k = f$ on B_k, $f_k = 0$ elsewhere (locally summable on $\mathbb{R}^n$). For a.e. $x_0 \in B_k$ the averages of f_k over $B(x_0,r)$, which equal those of f once $B(x_0,r) \subset B_k$, behave as in (1). A countable union of null sets is a null set.

(3) If f is continuous at x_0, then x_0 is a Lebesgue point: $⨍_{B(x_0,r)} |f - f(x_0)| \le \sup_{B(x_0,r)} |f - f(x_0)| \to 0$.

0.6 Gauss–Green Theorem and Green's Formulas

Boundaries and normals. Let $U \subset \mathbb{R}^n$ be open and bounded, $k \in \{1, 2, \dots\}$.

Definition 0.19. We say ∂U is C^k if for each point $x^0 \in \partial U$ there exist $r > 0$ and a C^k function $\gamma : \mathbb{R}^{n-1} \to \mathbb{R}$ such that—upon relabeling and reorienting the coordinate axes if necessary—we have

$$U \cap B(x^0, r) = \{x \in B(x^0, r) \mid x_n > \gamma(x_1, \dots, x_{n-1})\}.$$

Definition 0.20. (i) If ∂U is C^1, then along ∂U is defined the *outward pointing* unit normal vector field

$$\nu = (\nu^1, \dots, \nu^n).$$

The unit normal at any point $x^0 \in \partial U$ is $\nu(x^0) = \nu = (\nu_1, \dots, \nu_n)$.

(ii) Let $u \in C^1(\bar{U})$. We call

$$\frac{\partial u}{\partial \nu} := \nu \cdot Du$$

the (outward) normal derivative of u.

Example 0.21 (The two domains we need)**.** (a) *A ball.* For $U = B^\circ(x, r)$ the outward unit normal at $y \in \partial B(x, r)$ is $\nu(y) = \frac{y-x}{r}$, and $\frac{\partial u}{\partial \nu}(y) = Du(y) \cdot \frac{y-x}{r}$.

(b) *An annulus.* Let $0 < \varepsilon < R$ and $U = \{y \in \mathbb{R}^n : \varepsilon < |y| < R\}$. Its boundary consists of two spheres, $\partial U = \partial B(0, R) \cup \partial B(0, \varepsilon)$, both smooth. The outward unit normal of U is

$$\nu(y) = \frac{y}{R} \quad \text{on } \partial B(0, R), \qquad \nu(y) = -\frac{y}{\varepsilon} \quad \text{on } \partial B(0, \varepsilon).$$

On the outer sphere "outward from U" means away from the origin. On the inner sphere, leaving U means going *into the hole* $B(0, \varepsilon)$, that is, *towards* the origin. So on $\partial B(0, \varepsilon)$ the normal $\nu = -y/\varepsilon$ points *inward* with respect to the small ball $B(0, \varepsilon)$.

In the following results we assume U is a bounded, open subset of $\mathbb{R}^n$, and ∂U is C^1.

Theorem 0.22 (Gauss–Green Theorem)**.** *Suppose $u \in C^1(\bar{U})$. Then*

$$\int_U u_{x_i}\, dx = \int_{\partial U} u\nu^i\, dS \quad (i = 1, \dots, n).$$

Theorem 0.23 (Integration-by-parts formula)**.** *Let $u, v \in C^1(\bar{U})$. Then*

$$\int_U u_{x_i} v\, dx = -\int_U u v_{x_i}\, dx + \int_{\partial U} uv\nu^i\, dS \quad (i = 1, \dots, n). \tag{2}$$

Proof. Apply the Gauss–Green theorem to the product $uv \in C^1(\bar{U})$: since $(uv)_{x_i} = u_{x_i} v + u v_{x_i}$,

$$\int_U u_{x_i} v\, dx + \int_U u v_{x_i}\, dx = \int_{\partial U} uv\nu^i\, dS.$$

Rearranging gives (2). □

Theorem 0.24 (Green's formulas)**.** *Let $u, v \in C^2(\bar{U})$. Then*

(i) $\displaystyle\int_U \Delta u\, dx = \int_{\partial U} \frac{\partial u}{\partial \nu}\, dS,$

(ii) $\displaystyle\int_U Dv \cdot Du\, dx = -\int_U u\Delta v\, dx + \int_{\partial U} \frac{\partial v}{\partial \nu} u\, dS,$

(iii) $\displaystyle\int_U u\Delta v - v\Delta u\, dx = \int_{\partial U} u\frac{\partial v}{\partial \nu} - v\frac{\partial u}{\partial \nu}\, dS.$

Proof. (i) Use (2) with u_{x_i} in place of u and $v \equiv 1$:

$$\int_U u_{x_i x_i}\,dx = \int_{\partial U} u_{x_i}\nu^i\,dS.$$

Summing over $i = 1, \dots, n$, the left side becomes $\int_U \Delta u\,dx$ and the right side becomes

$$\int_{\partial U} \sum_i \nu^i u_{x_i}\,dS = \int_{\partial U} \nu \cdot Du\,dS = \int_{\partial U} \frac{\partial u}{\partial \nu}\,dS.$$

(ii) Use (2) with v_{x_i} in place of u and u in place of v:

$$\int_U v_{x_i} u_{x_i}\,dx = -\int_U v_{x_i x_i} u\,dx + \int_{\partial U} v_{x_i}\nu^i u\,dS.$$

Summing over i gives $\int_U Dv \cdot Du\,dx = -\int_U u\Delta v\,dx + \int_{\partial U}(\nu \cdot Dv)u\,dS$, which is (ii).

(iii) Write (ii) with u and v interchanged and subtract it from (ii); the terms $\int_U Du \cdot Dv\,dx$ cancel. □

0.7 Convolution and Smoothing

We next introduce tools that will allow us to build smooth approximations to given functions. Recall that the convolution of f and g is $(f*g)(x) = \int f(x-y)g(y)\,dy$; the substitution $z = x-y$ shows $f * g = g * f$.

Lemma 0.25 (A smooth function with compact support)**.** *The function*

$$\psi(t) = \begin{cases} \exp\left(\dfrac{1}{t-1}\right) & \text{if } t < 1 \\ 0 & \text{if } t \geq 1 \end{cases}$$

belongs to $C^\infty(\mathbb{R})$*, and* $\psi > 0$ *on* $(-\infty, 1)$*.*

Proof. On $t < 1$ and on $t > 1$ the function is clearly C^∞; the only issue is the point $t = 1$. Write $s = \frac{1}{t-1}$; as $t \to 1^-$ we have $s \to -\infty$, and $\frac{ds}{dt} = -\frac{1}{(t-1)^2} = -s^2$.

Claim: for $t < 1$*,* $\psi^{(k)}(t) = P_k(s)e^s$ *for a polynomial* P_k*.* For $k = 0$, $P_0 = 1$. If it holds for k, then by the chain rule

$$\psi^{(k+1)}(t) = \frac{d}{ds}\big(P_k(s)e^s\big) \cdot \frac{ds}{dt} = -s^2\big(P_k'(s) + P_k(s)\big)e^s,$$

so $P_{k+1}(s) = -s^2(P_k'(s) + P_k(s))$ is again a polynomial.

Key limit: for every polynomial P, $P(s)e^s \to 0$ as $s \to -\infty$ (the exponential decays faster than any power grows; e.g. $|s|^j e^s = |s|^j / e^{|s|} \to 0$ by repeated L'Hôpital).

Induction: $\psi^{(k)}$ *exists on* $\mathbb{R}$*, is continuous, and* $\psi^{(k)}(t) = 0$ *for* $t \geq 1$. For $k = 0$: $\psi(t) = e^s \to 0 = \psi(1)$ as $t \to 1^-$. Assume it for k. For $t > 1$ clearly $\psi^{(k+1)}(t) = 0$. At $t = 1$, the right difference quotient of $\psi^{(k)}$ is 0, and the left one is

$$\frac{\psi^{(k)}(t) - \psi^{(k)}(1)}{t-1} = s\,P_k(s)e^s \to 0 \qquad (t \to 1^-),$$

because $sP_k(s)$ is a polynomial. Hence $\psi^{(k+1)}(1) = 0$. Finally $\psi^{(k+1)}(t) = P_{k+1}(s)e^s \to 0$ as $t \to 1^-$, so $\psi^{(k+1)}$ is continuous at 1. □

Definition 0.26 (Standard mollifier)**.** (i) Define $\eta \in C^\infty(\mathbb{R}^n)$ by

$$\eta(x) := \begin{cases} C \exp\left(\dfrac{1}{|x|^2 - 1}\right) & \text{if } |x| < 1 \\ 0 & \text{if } |x| \geq 1, \end{cases}$$

the constant $C > 0$ selected so that $\int_{\mathbb{R}^n} \eta \, dx = 1$.

(ii) For each $\varepsilon > 0$, set

$$\eta_\varepsilon(x) := \frac{1}{\varepsilon^n} \eta\left(\frac{x}{\varepsilon}\right).$$

We call η the *standard mollifier*. The functions η_ε are C^∞ and satisfy

$$\int_{\mathbb{R}^n} \eta_\varepsilon \, dx = 1, \qquad \operatorname{spt}(\eta_\varepsilon) \subset B(0, \varepsilon).$$

Why these properties hold.

- $\eta(x) = C\psi(|x|^2)$ with ψ from Lemma 0.25. Since $x \mapsto |x|^2 = x_1^2 + \cdots + x_n^2$ is a polynomial, η is a composition of C^∞ functions, so $\eta \in C^\infty(\mathbb{R}^n)$. Also $\eta > 0$ on $B^\circ(0,1)$ and $\eta = 0$ outside, so $0 < \int \eta \, dx < \infty$ (η is continuous with compact support), and C can be chosen.
- η_ε is η composed with the linear map $x \mapsto x/\varepsilon$, times a constant, hence C^∞.
- $\eta_\varepsilon(x) \neq 0$ iff $|x/\varepsilon| < 1$ iff $|x| < \varepsilon$; the closure of this set is $B(0,\varepsilon)$, so $\operatorname{spt} \eta_\varepsilon = B(0,\varepsilon)$.
- Substituting $x = \varepsilon z$, $dx = \varepsilon^n dz$: $\int \eta_\varepsilon(x) \, dx = \int \varepsilon^{-n} \eta(z) \, \varepsilon^n dz = \int \eta(z) \, dz = 1$.
- $\eta \geq 0$, $\eta_\varepsilon \geq 0$, and η is *radial*: $\eta(x)$ depends only on $|x|$. We write $\eta(x) = \eta(|x|)$, using the same letter for the function of one variable $r \mapsto C\psi(r^2)$ (see the Remark on notation before Theorem 6).

Definition 0.27 (Mollification)**.** If $f : U \to \mathbb{R}$ is locally integrable, define its *mollification*

$$f^\varepsilon := \eta_\varepsilon * f \quad \text{in } U_\varepsilon.$$

That is,

$$f^\varepsilon(x) = \int_U \eta_\varepsilon(x - y) f(y) \, dy = \int_{B(0,\varepsilon)} \eta_\varepsilon(y) f(x - y) \, dy$$

for $x \in U_\varepsilon$.

Why only in U_ε: $\eta_\varepsilon(x - y) \neq 0$ only for $y \in B(x, \varepsilon)$, and for $x \in U_\varepsilon$ the closed ball $B(x,\varepsilon)$ lies in U (Lemma 0.2(a)) and is compact, so f is integrable there and $f^\varepsilon(x)$ is finite. For x closer than ε to ∂U the ball would leave U, where f is not defined. The second formula comes from $z = x - y$.

Theorem 0.28 (Properties of mollifiers)**.** *(i)* $f^\varepsilon \in C^\infty(U_\varepsilon)$.

(ii) $f^\varepsilon \to f$ *a.e. as* $\varepsilon \to 0$.

(iii) If $f \in C(U)$*, then* $f^\varepsilon \to f$ *uniformly on compact subsets of* U.

(iv) If $1 \leq p < \infty$ *and* $f \in L^p_{\mathrm{loc}}(U)$*, then* $f^\varepsilon \to f$ *in* $L^p_{\mathrm{loc}}(U)$.

Proof. (i) This is Lemma 0.15 with $g = \eta_\varepsilon$ and $\rho = \varepsilon$: $\eta_\varepsilon \in C^\infty(\mathbb{R}^n)$ vanishes for $|z| > \varepsilon$, and $f \in L^1_{\mathrm{loc}}(U)$. Hence $f^\varepsilon \in C^\infty(U_\varepsilon)$ and

$$D^\alpha f^\varepsilon(x) = \int_U D^\alpha \eta_\varepsilon(x-y) f(y)\,dy \quad (x \in U_\varepsilon)$$

for each multiindex α. (The essential point, proved in that lemma, is that the difference quotients $\frac{1}{h}[\eta_\varepsilon(x+he_i-y) - \eta_\varepsilon(x-y)]$ converge to $\frac{\partial \eta_\varepsilon}{\partial x_i}(x-y)$ *uniformly* in y, so the limit can be taken inside the integral.)

A basic estimate. Since $\int \eta_\varepsilon(x-y)\,dy = 1$, we can write the constant $f(x)$ as $f(x) = \int_{B(x,\varepsilon)} \eta_\varepsilon(x-y) f(x)\,dy$. Subtracting, for $x \in U_\varepsilon$,

$$|f^\varepsilon(x) - f(x)| = \left| \int_{B(x,\varepsilon)} \eta_\varepsilon(x-y)\big[f(y) - f(x)\big]\,dy \right| \le \int_{B(x,\varepsilon)} \eta_\varepsilon(x-y)\,|f(y) - f(x)|\,dy, \quad (3)$$

using $\eta_\varepsilon \ge 0$.

(ii) By Lebesgue's Differentiation Theorem 0.17 (in its local form, Remark (2) after it), for a.e. $x \in U$,

$$\lim_{r \to 0} \fint_{B(x,r)} |f(y) - f(x)|\,dy = 0.$$

Fix such a point x. For ε small, $x \in U_\varepsilon$. Since $\eta_\varepsilon(x-y) = \varepsilon^{-n}\eta(\frac{x-y}{\varepsilon}) \le \varepsilon^{-n}\max\eta$, (3) gives

$$|f^\varepsilon(x) - f(x)| \le \frac{\max\eta}{\varepsilon^n} \int_{B(x,\varepsilon)} |f(y) - f(x)|\,dy = \alpha(n)\max\eta \fint_{B(x,\varepsilon)} |f(y) - f(x)|\,dy \to 0$$

as $\varepsilon \to 0$ (we multiplied and divided by $|B(x,\varepsilon)| = \alpha(n)\varepsilon^n$). Assertion (ii) follows.

(iii) Assume $f \in C(U)$ and let $K \subset U$ be compact. Since K is compact and ∂U is closed and disjoint from K, $\delta := \operatorname{dist}(K, \partial U) > 0$ (if $U = \mathbb{R}^n$ take $\delta = 1$). Let $W = \{y : \operatorname{dist}(y,K) \le \delta/2\}$. W is closed and bounded, hence compact, and $W \subset U$: if $\operatorname{dist}(y,K) \le \delta/2$ pick $k \in K$ with $|y-k| \le \delta/2 < \operatorname{dist}(k, \partial U)$ and use Lemma 0.2(a). So f is uniformly continuous on W: for $\eta' > 0$ there is $\sigma > 0$ with $|f(y) - f(x)| < \eta'$ whenever $x, y \in W$, $|x-y| \le \sigma$.

Now let $x \in K$ and $\varepsilon < \min(\delta/2, \sigma)$. Then $\operatorname{dist}(x, \partial U) \ge \delta > \varepsilon$, so $x \in U_\varepsilon$, and $B(x,\varepsilon) \subset W$. By (3),

$$|f^\varepsilon(x) - f(x)| \le \eta' \int_{B(x,\varepsilon)} \eta_\varepsilon(x-y)\,dy = \eta'.$$

This bound does not depend on $x \in K$, so $\sup_K |f^\varepsilon - f| \to 0$.

(iv) Assume $1 \le p < \infty$, $f \in L^p_{\mathrm{loc}}(U)$, and let $V \subset\subset U$ be open. As in (iii) (with $K = \bar{V}$), choose $\delta = \operatorname{dist}(V, \partial U) > 0$ and the compact set $W = \{y : \operatorname{dist}(y,V) \le \delta/2\} \subset U$. We must show $\|f^\varepsilon - f\|_{L^p(V)} \to 0$.

Step 1: the estimate $\|f^\varepsilon\|_{L^p(V)} \le \|f\|_{L^p(W)}$ for $\varepsilon < \delta/2$. Let $x \in V$; then $B(x,\varepsilon) \subset W$. If $p > 1$, let $q = \frac{p}{p-1}$ so $\frac{1}{p} + \frac{1}{q} = 1$, and split $\eta_\varepsilon = \eta_\varepsilon^{1/q}\eta_\varepsilon^{1/p}$. By Hölder's inequality,

$$\begin{aligned} |f^\varepsilon(x)| &\le \int_{B(x,\varepsilon)} \eta_\varepsilon^{1/q}(x-y)\,\eta_\varepsilon^{1/p}(x-y)|f(y)|\,dy \\ &\le \left(\int_{B(x,\varepsilon)} \eta_\varepsilon(x-y)\,dy \right)^{1/q} \left(\int_{B(x,\varepsilon)} \eta_\varepsilon(x-y)|f(y)|^p\,dy \right)^{1/p}. \end{aligned}$$

The first factor is 1. Raising to the power p:

$$|f^\varepsilon(x)|^p \le \int_W \eta_\varepsilon(x-y)|f(y)|^p\,dy.$$

(For $p = 1$ this is immediate from the definition.) Integrate over $x \in V$ and exchange the order of integration (Tonelli, the integrand is ≥ 0):

$$\int_V |f^\varepsilon|^p dx \leq \int_W |f(y)|^p \left(\int_V \eta_\varepsilon(x-y)\, dx \right) dy \leq \int_W |f(y)|^p dy,$$

since $\int_V \eta_\varepsilon(x-y)\, dx \leq \int_{\mathbb{R}^n} \eta_\varepsilon(x-y)\, dx = 1$.

Step 2: approximation by a continuous function. Fix $\delta' > 0$. By density (Fact 0.16(5)) choose g continuous with compact support in W° such that $\|f - g\|_{L^p(W)} < \delta'$; extend g by zero, so $g \in C(U)$. Mollification is linear, $f^\varepsilon - g^\varepsilon = (f-g)^\varepsilon$, so Step 1 applied to $f - g$ gives $\|f^\varepsilon - g^\varepsilon\|_{L^p(V)} \leq \|f - g\|_{L^p(W)} < \delta'$. By the triangle inequality,

$$\|f^\varepsilon - f\|_{L^p(V)} \leq \|f^\varepsilon - g^\varepsilon\|_{L^p(V)} + \|g^\varepsilon - g\|_{L^p(V)} + \|g - f\|_{L^p(V)} \leq 2\delta' + \|g^\varepsilon - g\|_{L^p(V)}.$$

By (iii), $g^\varepsilon \to g$ uniformly on the compact set $\bar{V}$, so $\|g^\varepsilon - g\|_{L^p(V)} \leq |V|^{1/p} \sup_V |g^\varepsilon - g| \to 0$. Therefore $\limsup_{\varepsilon \to 0} \|f^\varepsilon - f\|_{L^p(V)} \leq 2\delta'$, and since $\delta' > 0$ was arbitrary, the limit is 0. □

0.8 Inverse and Implicit Function Theorems

Let $U \subset \mathbb{R}^n$ be an open set and suppose $\mathbf{f} : U \to \mathbb{R}^n$ is C^1, $\mathbf{f} = (f^1, \dots, f^n)$. Assume $x_0 \in U$, $z_0 = \mathbf{f}(x_0)$. We write

$$D\mathbf{f} = \begin{pmatrix} f^1_{x_1} & \cdots & f^1_{x_n} \\ & \ddots & \\ f^n_{x_1} & \cdots & f^n_{x_n} \end{pmatrix}_{n \times n} = \text{gradient matrix of } \mathbf{f}.$$

Definition 0.29.

$$J\mathbf{f} = \text{Jacobian of } \mathbf{f} = |\det D\mathbf{f}| = \left| \frac{\partial(f^1, \dots, f^n)}{\partial(x_1, \dots, x_n)} \right|.$$

Theorem 0.30 (Inverse Function Theorem)**.** *Assume* $\mathbf{f} \in C^1(U; \mathbb{R}^n)$ *and*

$$J\mathbf{f}(x_0) \neq 0.$$

Then there exist an open set $V \subset U$*, with* $x_0 \in V$*, and an open set* $W \subset \mathbb{R}^n$*, with* $z_0 \in W$*, such that*

(i) the mapping $\mathbf{f} : V \to W$ *is one-to-one and onto, and*

(ii) the inverse function $\mathbf{f}^{-1} : W \to V$ *is* C^1*.*

(iii) If $\mathbf{f} \in C^k$*, then* $\mathbf{f}^{-1} \in C^k$ $(k = 2, \dots)$*.*

Let n, m be positive integers. We write a typical point in $\mathbb{R}^{n+m}$ as

$$(x, y) = (x_1, \dots, x_n, y_1, \dots, y_m)$$

for $x \in \mathbb{R}^n$, $y \in \mathbb{R}^m$. Let $U \subset \mathbb{R}^{n+m}$ be an open set and suppose $\mathbf{f} : U \to \mathbb{R}^m$ is C^1, $\mathbf{f} = (f^1, \dots, f^m)$. Assume $(x_0, y_0) \in U$, $z_0 = \mathbf{f}(x_0, y_0)$. We write

$$D\mathbf{f} = \begin{pmatrix} f^1_{x_1} & \cdots & f^1_{x_n} & f^1_{y_1} & \cdots & f^1_{y_m} \\ & \ddots & & & \ddots & \\ f^m_{x_1} & \cdots & f^m_{x_n} & f^m_{y_1} & \cdots & f^m_{y_m} \end{pmatrix}_{m \times (n+m)} = (D_x\mathbf{f}, D_y\mathbf{f}) = \text{gradient matrix of } \mathbf{f}.$$

Definition 0.31.

$$J_y\mathbf{f} = |\det D_y\mathbf{f}| = \left| \frac{\partial(f^1, \dots, f^m)}{\partial(y_1, \dots, y_m)} \right|.$$

Theorem 0.32 (Implicit Function Theorem)**.** *Assume* $\mathbf{f} \in C^1(U; \mathbb{R}^m)$ *and*

$$J_y \mathbf{f}(x_0, y_0) \neq 0.$$

Then there exist an open set $V \subset U$*, with* $(x_0, y_0) \in V$*, an open set* $W \subset \mathbb{R}^n$*, with* $x_0 \in W$*, and a* C^1 *mapping* $\mathbf{g} : W \to \mathbb{R}^m$ *such that*

(i) $\mathbf{g}(x_0) = y_0$*,*

(ii) $\mathbf{f}(x, \mathbf{g}(x)) = z_0 \ (x \in W)$*,*

and

(iii) if $(x, y) \in V$ *and* $\mathbf{f}(x, y) = z_0$*, then* $y = \mathbf{g}(x)$*.*

(iv) If $\mathbf{f} \in C^k$*, then* $\mathbf{g} \in C^k \ (k = 2, \dots)$*.*

The function $\mathbf{g}$ *is* implicitly defined *near* x_0 *by the equation* $\mathbf{f}(x, y) = z_0$*.*

1 Method of Characteristics

1.1 Introduction and Linear Equations

1.1.1 Introduction

The method of characteristics is a powerful technique for solving first-order partial differential equations (PDEs). The fundamental idea is to reduce the PDE to a system of ordinary differential equations (ODEs) along specific curves known as *characteristic curves*. By solving these ODEs, we can reconstruct the solution surface $u(x, y)$.

1.1.2 Linear Equations

Consider the general linear first-order PDE in two variables:

$$a(x, y)u_x + b(x, y)u_y = c(x, y)u + d(x, y) \tag{4}$$

We wish to find a solution $u(x, y)$ that satisfies a given initial condition on a curve Γ.

Step-by-step Method:

1. **Parameterize the initial curve:** Let the initial data be given on a curve Γ, parameterized by s:
$$x = x_0(s), \quad y = y_0(s), \quad u = u_0(s)$$

2. **Form the Characteristic Equations:** We look for characteristic curves $(x(\tau, s), y(\tau, s))$ parameterized by τ, emanating from the initial curve at $\tau = 0$. The characteristic ODEs are:
$$\frac{dx}{d\tau} = a(x, y), \quad \frac{dy}{d\tau} = b(x, y)$$
Along these curves, the PDE simplifies to an ODE for $u(\tau, s)$:
$$\frac{du}{d\tau} = c(x, y)u + d(x, y)$$

3. **Solve the ODEs:** Solve the system of ODEs subject to the initial conditions at $\tau = 0$:
$$x(0, s) = x_0(s), \quad y(0, s) = y_0(s), \quad u(0, s) = u_0(s)$$

4. **Invert the Coordinates:** Solve for τ and s in terms of x and y. Substitute back to find $u(x, y)$.

1.2 Quasilinear Equations

A quasilinear equation is linear in the derivatives u_x and u_y, but the coefficients can depend on u:

$$a(x, y, u)u_x + b(x, y, u)u_y = c(x, y, u) \tag{5}$$

1.2.1 Burgers' Equation

A classic example is the inviscid Burgers' equation:

$$u_t + uu_x = 0$$

with initial data $u(x, 0) = h(x)$. The characteristic equations are:

$$\frac{dx}{dt} = u, \quad \frac{du}{dt} = 0$$

This implies that u is constant along characteristics, and the characteristics are straight lines with slope (velocity) equal to the initial value $h(x)$. The solution implicitly is $u = h(x - ut)$.

Shock and Rarefaction Waves:

- **Shock Wave:** If $h(x)$ is decreasing, the characteristics will eventually intersect. At the intersection, the solution becomes multi-valued and a singularity develops. We define a *shock wave* as a propagating discontinuity that satisfies a specific physical condition (Rankine-Hugoniot jump condition) to remain a valid weak solution.
- **Rarefaction Wave:** If $h(x)$ has a discontinuous jump where it increases, a gap opens between characteristics. This region is filled by a continuous fan of characteristics known as a *rarefaction wave*, smoothing out the initial discontinuity over time.

1.2.2 Definition of Integral Surface

An *integral surface* for the PDE is the geometric surface $z = u(x, y)$ in $\mathbb{R}^3$ that represents the solution. The PDE $a(x, y, u)u_x + b(x, y, u)u_y = c(x, y, u)$ states that the vector field (a, b, c) is always tangent to the integral surface. Therefore, the integral surface is constructed by taking the union of all characteristic curves passing through the initial data curve.

1.2.3 Theorem and Proof (Application of the Inverse Function Theorem)

Theorem (Cauchy Problem): Suppose a, b, c and the initial curve Γ are C^1. If the transversality condition holds:

$$a(x_0(s), y_0(s), u_0(s))y_0'(s) - b(x_0(s), y_0(s), u_0(s))x_0'(s) \neq 0$$

then there exists a unique C^1 solution $u(x, y)$ in a neighborhood of the initial curve.

Proof in Steps:

1. **Characteristics:** Construct the system of ODEs: $\frac{dx}{d\tau} = a$, $\frac{dy}{d\tau} = b$, $\frac{du}{d\tau} = c$ with initial data at $\tau = 0$ parameterized by s. By the Picard-Lindelöf theorem, there exist unique solutions $x(\tau, s), y(\tau, s), u(\tau, s)$.
2. **Applying the Inverse Function Theorem:** To define u as a function of x and y, we need to invert the mapping $(\tau, s) \mapsto (x, y)$. By the Inverse Function Theorem (Theorem 0.30), this mapping is locally invertible near $\tau = 0$ (with a C^1 inverse) if the Jacobian determinant is non-zero:

$$J = \det \begin{pmatrix} \frac{\partial x}{\partial \tau} & \frac{\partial x}{\partial s} \\ \frac{\partial y}{\partial \tau} & \frac{\partial y}{\partial s} \end{pmatrix} = \frac{\partial x}{\partial \tau}\frac{\partial y}{\partial s} - \frac{\partial y}{\partial \tau}\frac{\partial x}{\partial s} \neq 0$$

3. **Checking the Requirement:** At $\tau = 0$, we have $\frac{\partial x}{\partial \tau} = a$, $\frac{\partial y}{\partial \tau} = b$, $\frac{\partial x}{\partial s} = x_0'(s)$, and $\frac{\partial y}{\partial s} = y_0'(s)$. Therefore, the Jacobian is exactly the transversality condition: $ay_0' - bx_0'$. Since this is non-zero by assumption, the Inverse Function Theorem guarantees we can locally find C^1 functions $\tau(x, y)$ and $s(x, y)$.

4. **Conclusion:** Substituting these back, we get $u(x, y) = u(\tau(x, y), s(x, y))$, which is the unique C^1 solution.

1.3 General First-Order Equation in Two Variables

Consider the fully nonlinear equation:

$$F(x, y, u, p, q) = 0$$

where $p = u_x$ and $q = u_y$.

1.3.1 Idea of the Proof in Steps

To solve this, we cannot simply use the vector field (a, b, c) as in the quasilinear case because the coefficients now depend nonlinearly on p and q. The idea is to track not just the spatial coordinates (x, y) and value u, but also the derivatives p and q along the characteristic curve.

1. Differentiate the PDE with respect to x and y to obtain relations for p_x, p_y, q_x, q_y.

2. Construct a system of five ODEs for the five variables (x, y, u, p, q). These are called the *Characteristic Strip* equations.

3. Solve this system subject to initial conditions. Ensure the initial data for $p_0(s)$ and $q_0(s)$ are compatible with $F = 0$ and the strip condition $u_0'(s) = p_0(s)x_0'(s) + q_0(s)y_0'(s)$.

1.3.2 Step-by-Step Calculation

Differentiating $F(x, y, u, p, q) = 0$ with respect to x:

$$F_x + F_u u_x + F_p p_x + F_q q_x = 0 \implies F_x + pF_u + p_x F_p + p_y F_q = 0$$

(using $q_x = p_y$). We choose our parameter τ such that:

$$\frac{dx}{d\tau} = F_p, \quad \frac{dy}{d\tau} = F_q$$

Then along this curve, $\frac{dp}{d\tau} = p_x\frac{dx}{d\tau} + p_y\frac{dy}{d\tau} = p_xF_p + p_yF_q = -(F_x + pF_u)$. Similarly, $\frac{dq}{d\tau} = -(F_y + qF_u)$. The change in u is $\frac{du}{d\tau} = p\frac{dx}{d\tau} + q\frac{dy}{d\tau} = pF_p + qF_q$. These form the characteristic equations.

1.3.3 Geometric Explanation of the Monge Cone

At any point (x_0, y_0, u_0), the PDE restricts the possible tangent planes $(p, q, -1)$ to the solution surface via $F(x_0, y_0, u_0, p, q) = 0$. Since $F = 0$ generally defines a 1-parameter family of planes in $\mathbb{R}^3$, the envelope of all these possible tangent planes forms a cone, called the **Monge Cone**. The integral surface representing the solution must be tangent to a Monge cone at every point. The method of characteristics traces the paths that lie exactly along the contact lines between the integral surface and these Monge cones.

1.4 First-Order Equation in Several Variables

Here, we generalize the ideas to n independent variables $x = (x_1, \ldots, x_n)$.

1.4.1 Linear First-Order Equation

The equation is $\sum_{i=1}^{n} a_i(x)u_{x_i} = c(x)u + d(x)$. The characteristic equations simply extend to n spatial ODEs: $\frac{dx_i}{d\tau} = a_i(x)$, along which $\frac{du}{d\tau} = c(x)u + d(x)$. The initial data is given on an $(n-1)$-dimensional hypersurface, and the solution is reconstructed by tracing characteristics from this hypersurface.

1.4.2 Quasilinear Equation

For $\sum_{i=1}^{n} a_i(x,u)u_{x_i} = c(x,u)$, the characteristic equations are $\frac{dx_i}{d\tau} = a_i(x,u)$ and $\frac{du}{d\tau} = c(x,u)$. The geometry is identical to the 2D case, where the vector field $(a_1, \ldots, a_n, c)$ in $\mathbb{R}^{n+1}$ must be tangent to the n-dimensional integral surface. The Inverse Function Theorem (Theorem 0.30) is again used, after verifying that the characteristics do not run parallel to the initial hypersurface.

1.4.3 General Nonlinear Equation

For $F(x,u,Du) = 0$ where $Du = (p_1, \ldots, p_n)$, we form the characteristic strip in $2n+1$ dimensions (x,u,p). The equations generalize gracefully:

$$\frac{dx_i}{d\tau} = F_{p_i}, \quad \frac{du}{d\tau} = \sum_{i=1}^{n} p_i F_{p_i}, \quad \frac{dp_i}{d\tau} = -(F_{x_i} + p_i F_u)$$

The initial hypersurface must be augmented with compatible values for all partial derivatives p_i, satisfying both the PDE itself and the geometric strip condition on the hypersurface. Once again, by solving this extended system of ODEs, we can construct the unique local solution.

2 Laplace's Equation

2.1 Definitions

The **Laplace equation** is given by:

$$\Delta u = 0$$

where $\Delta u = \sum_{i=1}^{n} u_{x_i x_i}$ is the Laplacian of u. A C^2 function u satisfying Laplace's equation is called a **harmonic function**. If a source term f is present, the equation becomes **Poisson's equation**:

$$-\Delta u = f$$

2.2 Rotational Invariance of the Laplacian

The Laplacian operator is invariant under rotations of the coordinate system.

2.2.1 2D Case with 2×2 Matrix

Let $u(x,y)$ be a function in $\mathbb{R}^2$. Consider a rotation matrix:

$$O = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

Let new coordinates be $(x',y')^T = O(x,y)^T$. Since $O^T O = I$, we have $(x,y)^T = O^T(x',y')^T$, meaning:

$$x = x'\cos\theta + y'\sin\theta$$
$$y = -x'\sin\theta + y'\cos\theta$$

Let $v(x', y') = u(x(x', y'), y(x', y'))$. Using the chain rule:

$$v_{x'} = u_x \frac{\partial x}{\partial x'} + u_y \frac{\partial y}{\partial x'} = u_x \cos\theta - u_y \sin\theta$$

$$v_{y'} = u_x \frac{\partial x}{\partial y'} + u_y \frac{\partial y}{\partial y'} = u_x \sin\theta + u_y \cos\theta$$

Taking second derivatives:

$$v_{x'x'} = (u_{xx}\cos\theta - u_{yx}\sin\theta)\cos\theta - (u_{xy}\cos\theta - u_{yy}\sin\theta)\sin\theta$$

$$v_{x'x'} = u_{xx}\cos^2\theta - 2u_{xy}\sin\theta\cos\theta + u_{yy}\sin^2\theta$$

$$v_{y'y'} = (u_{xx}\sin\theta + u_{yx}\cos\theta)\sin\theta + (u_{xy}\sin\theta + u_{yy}\cos\theta)\cos\theta$$

$$v_{y'y'} = u_{xx}\sin^2\theta + 2u_{xy}\sin\theta\cos\theta + u_{yy}\cos^2\theta$$

Adding them together:

$$v_{x'x'} + v_{y'y'} = u_{xx}(\cos^2\theta + \sin^2\theta) + u_{yy}(\sin^2\theta + \cos^2\theta) = u_{xx} + u_{yy}$$

Thus, $\Delta_{x'} v = \Delta_x u$.

2.2.2 $\mathbb{R}^n$ Case

Let $y = Ox$ where O is an $n \times n$ orthogonal matrix ($O^T O = I$, or $\sum_k O_{ki} O_{kj} = \delta_{ij}$). Define $v(x) = u(Ox)$. Using index notation, $y_i = \sum_j O_{ij} x_j$, so $\frac{\partial y_i}{\partial x_k} = O_{ik}$. By the chain rule:

$$v_{x_k} = \sum_i u_{y_i} \frac{\partial y_i}{\partial x_k} = \sum_i u_{y_i} O_{ik}$$

$$v_{x_k x_k} = \sum_i \sum_j u_{y_i y_j} \frac{\partial y_j}{\partial x_k} O_{ik} = \sum_{i,j} u_{y_i y_j} O_{jk} O_{ik}$$

Summing over k to form the Laplacian:

$$\Delta v = \sum_k v_{x_k x_k} = \sum_{i,j} u_{y_i y_j} \left(\sum_k O_{ik} O_{jk} \right) = \sum_{i,j} u_{y_i y_j} \delta_{ij} = \sum_i u_{y_i y_i} = \Delta u$$

This rotational invariance suggests that we should look for radial solutions to Laplace's equation.

2.3 Derivation of the Fundamental Solution

We seek a solution $u(x) = v(r)$ where $r = |x| = (x_1^2 + \cdots + x_n^2)^{1/2}$.

2.3.1 Radial Formula for Laplacian

First, observe $\frac{\partial r}{\partial x_i} = \frac{1}{2}(x_1^2 + \cdots + x_n^2)^{-1/2} 2x_i = \frac{x_i}{r}$. By the chain rule:

$$u_{x_i} = v'(r) \frac{\partial r}{\partial x_i} = v'(r) \frac{x_i}{r}$$

Now we compute the second derivative:

$$u_{x_i x_i} = \frac{\partial}{\partial x_i} \left(v'(r) \frac{x_i}{r} \right) = v''(r) \left(\frac{x_i}{r} \right)^2 + v'(r) \left(\frac{1 \cdot r - x_i (x_i / r)}{r^2} \right)$$

$$u_{x_i x_i} = v''(r)\frac{x_i^2}{r^2} + v'(r)\left(\frac{1}{r} - \frac{x_i^2}{r^3}\right)$$

Summing over $i = 1 \dots n$:

$$\Delta u = \sum_{i=1}^{n} u_{x_i x_i} = v''(r)\frac{\sum x_i^2}{r^2} + v'(r)\left(\frac{n}{r} - \frac{\sum x_i^2}{r^3}\right)$$

Since $\sum x_i^2 = r^2$, we obtain the radial Laplacian:

$$\Delta u = v''(r) + \frac{n-1}{r}v'(r) = 0$$

2.3.2 Solving the ODE

Let $w = v'$. The ODE becomes $w' + \frac{n-1}{r}w = 0$.

$$\frac{w'}{w} = \frac{1-n}{r} \implies \ln|w| = (1-n)\ln r + C \implies w(r) = \frac{c_1}{r^{n-1}}$$

So $v'(r) = \frac{c_1}{r^{n-1}}$. Integrating this, we have two cases based on dimension n:

- **For** $n = 2$: $v(r) = c_1 \ln r + c_2$.
- **For** $n \geq 3$: $v(r) = \frac{c_1}{(2-n)}\frac{1}{r^{n-2}} + c_2$.

2.3.3 Defining the Fundamental Solution

We choose the constants to normalize the total flux through a sphere to 1. The **fundamental solution** $\Phi(x)$ is defined as:

$$\Phi(x) = \begin{cases} -\frac{1}{2\pi}\ln|x| & (n=2) \\ \frac{1}{n(n-2)\alpha(n)}\frac{1}{|x|^{n-2}} & (n \geq 3) \end{cases}$$

where $\alpha(n)$ is the volume of the unit ball in $\mathbb{R}^n$.

2.3.4 Estimates of Derivatives

Let us calculate $D\Phi$ and $D^2\Phi$. For $n \geq 3$:

$$\Phi_{x_i} = \frac{1}{n(n-2)\alpha(n)}(2-n)|x|^{1-n}\frac{x_i}{|x|} = -\frac{1}{n\alpha(n)}\frac{x_i}{|x|^n}$$

Thus, $|D\Phi(x)| \leq \frac{C}{|x|^{n-1}}$. For the second derivative:

$$\Phi_{x_i x_j} = -\frac{1}{n\alpha(n)}\left(\frac{\delta_{ij}}{|x|^n} - n\frac{x_i x_j}{|x|^{n+2}}\right)$$

Thus, $|D^2\Phi(x)| \leq \frac{C}{|x|^n}$.

2.4 Polar Coordinates, Volume, and Integrability

2.4.1 Polar Coordinates in Integration

For radial functions, it is highly useful to integrate in spherical coordinates. The volume of a ball $B(0,R)$ is $V_n(R) = \alpha(n)R^n$. The surface area of the sphere $\partial B(0,R)$ is denoted $S_n(R)$. Since the ball is the union of spheres of radius r from 0 to R:

$$V_n(R) = \int_0^R S_n(r)dr$$

Assuming $S_n(r) = cr^{n-1}$, we get $\alpha(n)R^n = \frac{c}{n}R^n$, so $c = n\alpha(n)$. Thus, the surface area is $n\alpha(n)R^{n-1}$. For any integrable function f, we can write:

$$\int_{B(0,R)} f(x)dx = \int_0^R \left(\int_{\partial B(0,r)} f dS \right) dr$$

The precise statements (polar coordinates, balls with any center, scaling to the unit ball, radial functions) are collected in Fact 0.5, Theorem 0.6 and Corollary 0.7.

2.4.2 Local Integrability Lemma

We need to understand when $1/|x|^\gamma$ is locally integrable around the singularity at $x = 0$.

$$\int_{B(0,1)} \frac{1}{|x|^\gamma} dx = \int_0^1 \left(\int_{\partial B(0,r)} \frac{1}{r^\gamma} dS \right) dr$$

$$= \int_0^1 \frac{1}{r^\gamma} \left(n\alpha(n) r^{n-1} \right) dr = n\alpha(n) \int_0^1 r^{n-1-\gamma} dr$$

This integral converges if and only if the exponent $n - 1 - \gamma > -1$, which means $\gamma < n$. (The complete statement, including the behaviour at infinity, is Lemma 0.10.)

Conclusion: - The fundamental solution $\Phi(x) \sim \frac{1}{|x|^{n-2}}$ is locally integrable because $n-2 < n$. - $D\Phi \sim \frac{1}{|x|^{n-1}}$ is locally integrable because $n-1 < n$. - $D^2\Phi \sim \frac{1}{|x|^n}$ is **NOT** locally integrable because $n \not< n$.

2.5 Poisson's Equation and Convolution

To solve $-\Delta u = f$, we use the fundamental solution.

2.5.1 Definition of Convolution

The convolution of two functions f and g is defined as:

$$(f * g)(x) = \int_{\mathbb{R}^n} f(x-y)g(y)dy$$

2.5.2 Commutativity of Convolution

We prove $(f * g)(x) = (g * f)(x)$. **In 1D:**

$$\int_{-\infty}^{\infty} f(x-y)g(y)dy$$

Let $z = x - y$. Then $y = x - z$ and $dy = -dz$. As $y \to \infty, z \to -\infty$ and $y \to -\infty, z \to \infty$.

$$= \int_{\infty}^{-\infty} f(z)g(x-z)(-dz) = \int_{-\infty}^{\infty} g(x-z)f(z)dz = (g * f)(x)$$

In $\mathbb{R}^n$:

$$\int_{\mathbb{R}^n} f(x-y)g(y)dy$$

Let $z = x - y$. The Jacobian determinant of this transformation is $\det(-I) = (-1)^n$. The change of variables formula for multiple integrals requires the absolute value of the Jacobian, which is $|(-1)^n| = 1$. Thus $dy = dz$.

$$= \int_{\mathbb{R}^n} f(z)g(x-z)dz = (g * f)(x)$$

2.5.3 The Difficulty

We propose $u = \Phi * f$. If we apply the Laplacian directly:

$$\Delta u(x) = \Delta \int_{\mathbb{R}^n} \Phi(x-y)f(y)dy = \int_{\mathbb{R}^n} \Delta_x \Phi(x-y)f(y)dy$$

However, $\Delta\Phi = 0$ everywhere except at $x = y$ where it is undefined. We cannot simply pass the Laplacian inside because $|D^2\Phi(x)| \sim 1/|x|^n$, which is not integrable near 0. A more delicate proof is required.

2.6 Compact Support and Properties of Convolution

2.6.1 Compact Support

A function f has **compact support** if there exists a bounded set outside of which f is identically zero. The space of continuous functions with compact support is denoted $C_c(\mathbb{R}^n)$. (See Definition 0.3(4) for the support spt f, and Definition 0.8 and Example 0.9 for local integrability.)

2.6.2 Support of Convolution

Theorem: If f is continuous and compactly supported, and g is locally integrable, then $(f * g)(x)$ is well-defined everywhere. If both have compact support, then $f*g$ has compact support. *Step-by-step Proof:* Fix x. The integrand $f(x-y)g(y)$ is non-zero only when $x - y \in \text{supp}(f)$ and $y \in \text{supp}(g)$. If f has compact support, say $f(z) = 0$ for $|z| > R$, then $f(x-y)$ is non-zero only for $|x-y| \leq R$. Thus, the integration over y is restricted to a bounded ball $B(x, R)$. Since g is locally integrable, the integral over this bounded set is finite, so $(f * g)(x)$ exists. If both have compact support, say $\text{supp}(f) \subseteq B(0, R_1)$ and $\text{supp}(g) \subseteq B(0, R_2)$, then for the integrand to be non-zero, we need $|x - y| \leq R_1$ and $|y| \leq R_2$. By triangle inequality:

$$|x| = |x - y + y| \leq |x - y| + |y| \leq R_1 + R_2$$

So if $|x| > R_1 + R_2$, the integral is strictly zero. Thus, the convolution has compact support contained in $B(0, R_1 + R_2)$.

2.6.3 Smoothness of Convolution

(A more general version, which allows g to be merely locally integrable, is Lemma 0.15.)

Theorem: If $f \in C^2(\mathbb{R}^n)$ and has compact support, and g is continuous, then $u = f * g \in C^2(\mathbb{R}^n)$. *Step-by-step Proof:* We write $u(x) = \int_{\mathbb{R}^n} f(x-y)g(y)dy$. We want to show $\frac{\partial u}{\partial x_i} = \int_{\mathbb{R}^n} f_{x_i}(x-y)g(y)dy$. Consider the difference quotient:

$$\frac{u(x + he_i) - u(x)}{h} = \int_{\mathbb{R}^n} \frac{f(x + he_i - y) - f(x-y)}{h} g(y)dy$$

By the Mean Value Theorem, $\frac{f(x+he_i-y)-f(x-y)}{h} = f_{x_i}(x+\theta he_i-y)$ for some $\theta \in (0,1)$. Because f_{x_i} is continuous and has compact support, it is *uniformly continuous*. This means as $h \to 0$, $f_{x_i}(x+\theta he_i-y)$ converges to $f_{x_i}(x-y)$ **uniformly** in y. Because the convergence is uniform and the domain of integration is effectively bounded (since f has compact support), we can legitimately pass the limit inside the integral to obtain:

$$\frac{\partial u}{\partial x_i} = \int_{\mathbb{R}^n} f_{x_i}(x-y)g(y)dy$$

Applying the exact same argument to the first derivative, we get:

$$\frac{\partial^2 u}{\partial x_i \partial x_j} = \int_{\mathbb{R}^n} f_{x_i x_j}(x-y)g(y)dy$$

Alternative proof using Dominated Convergence Theorem (DCT): Let $F_n(y) = \frac{f(x+h_n e_i-y)-f(x-y)}{h_n} g(y)$. $F_n(y) \to f_{x_i}(x-y)g(y)$ point-wise as $h_n \to 0$. Since $f \in C^2$ and compactly supported, its derivative is globally bounded: $|f_{x_i}| \le M$. Thus by MVT, $|F_n(y)| \le M|g(y)|\chi_K(y)$ where K is a compact set slightly larger than the support of $f(x-\cdot)$. Since g is continuous on the compact set K, it is bounded, so $M|g(y)|\chi_K(y)$ is Lebesgue integrable. By the **Dominated Convergence Theorem**, we can interchange the limit and the integral:

$$\lim_{h_n \to 0} \int_{\mathbb{R}^n} F_n(y)dy = \int_{\mathbb{R}^n} \lim_{h_n \to 0} F_n(y)dy = \int_{\mathbb{R}^n} f_{x_i}(x-y)g(y)dy$$

Thus $u \in C^2$.

2.7 Solving Poisson's Equation

Recall that the naive computation $\Delta u(x) = \int_{\mathbb{R}^n} \Delta_x \Phi(x-y)f(y)\,dy = 0$ is wrong, because $D^2\Phi(x-y)$ is not integrable near the singularity $y = x$. We must proceed more carefully in calculating Δu.

Let us for simplicity now assume $f \in C_c^2(\mathbb{R}^n)$; that is, f is twice continuously differentiable, with compact support. We consider the convolution

$$u(x) = \int_{\mathbb{R}^n} \Phi(x-y)f(y)\,dy = \begin{cases} -\dfrac{1}{2\pi}\displaystyle\int_{\mathbb{R}^2} \log(|x-y|)f(y)\,dy & (n=2) \\ \dfrac{1}{n(n-2)\alpha(n)}\displaystyle\int_{\mathbb{R}^n} \dfrac{f(y)}{|x-y|^{n-2}}\,dy & (n \ge 3). \end{cases} \tag{6}$$

Here log denotes the natural logarithm. Since f is continuous with compact support and Φ is locally integrable, $u(x)$ is well defined for every $x \in \mathbb{R}^n$ (by the theorem on the support of convolutions above).

Tools used. The proof uses: the volumes and areas of balls and spheres (Fact 0.5); averages (Definition 0.4); polar coordinates (Corollary 0.7); differentiating a convolution (Lemma 0.15); the outward normal and the annulus (Definition 0.20, Example 0.21); Green's formulas (Theorem 0.24). We also use the notation $|D^2 f| = \left(\sum_{i,j} |f_{x_i x_j}|^2\right)^{1/2}$, $\|D^2 f\|_{L^\infty(\mathbb{R}^n)} = \sup_x |D^2 f(x)|$ and $\|Df\|_{L^\infty(\mathbb{R}^n)} = \sup_x |Df(x)|$; for $f \in C_c^2(\mathbb{R}^n)$ these suprema are finite, since a continuous function with compact support is bounded.

2.7.1 Preliminary Computations with Φ

(a) The gradient of Φ in every dimension. We computed Φ_{x_i} for $n \ge 3$. For $n = 2$, $\Phi(y) = -\frac{1}{2\pi}\log|y|$ and $\frac{\partial}{\partial y_i}\log|y| = \frac{1}{|y|}\cdot\frac{y_i}{|y|}$, so $\Phi_{y_i} = -\frac{1}{2\pi}\frac{y_i}{|y|^2}$. Since $\alpha(2) = \pi$, we have $2\pi = 2\alpha(2) = n\alpha(n)$. Therefore for all $n \ge 2$:

$$D\Phi(y) = -\frac{1}{n\alpha(n)}\frac{y}{|y|^n} \qquad (y \ne 0). \tag{7}$$

(b) Φ is harmonic away from the origin. Since $\Phi(y) = v(|y|)$ with v solving $v'' + \frac{n-1}{r}v' = 0$, the radial formula for the Laplacian gives $\Delta\Phi(y) = 0$ for all $y \neq 0$.

(c) Integrals of $|\Phi|$ over balls.

Lemma 2.1. *For every $\rho > 0$, $\int_{B(0,\rho)} |\Phi(y)|\,dy < \infty$; in particular $\Phi \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$. Moreover, for $0 < \varepsilon \le e^{-1/2}$,*

$$\int_{B(0,\varepsilon)} |\Phi(y)|\,dy \le \begin{cases} \varepsilon^2 |\log\varepsilon| & (n=2) \\ \dfrac{\varepsilon^2}{2(n-2)} & (n \ge 3). \end{cases}$$

Proof. $|\Phi|$ is radial, so we use Corollary 0.7(c). Case $n \ge 3$. Here $|\Phi(y)| = \frac{1}{n(n-2)\alpha(n)} r^{2-n}$ with $r = |y|$, and

$$\int_{B(0,\rho)} |\Phi|\,dy = n\alpha(n) \int_0^\rho \frac{r^{2-n}}{n(n-2)\alpha(n)}\, r^{n-1}\,dr = \frac{1}{n-2}\int_0^\rho r\,dr = \frac{\rho^2}{2(n-2)} < \infty.$$

(This is also Lemma 0.10(iv) with $m = n-2 < n$.) Taking $\rho = \varepsilon$ gives the stated bound.

Case $n = 2$. Here $|\Phi(y)| = \frac{1}{2\pi}|\log r|$ and $n\alpha(n) = 2\pi$, so

$$\int_{B(0,\rho)} |\Phi|\,dy = 2\pi \int_0^\rho \frac{|\log r|}{2\pi}\, r\,dr = \int_0^\rho r|\log r|\,dr.$$

The integrand $r|\log r|$ is continuous on $(0,\rho]$ and tends to 0 as $r \to 0^+$. Indeed, by L'Hôpital's rule,

$$\lim_{r\to 0^+} r\log r = \lim_{r\to 0^+} \frac{\log r}{1/r} = \lim_{r\to 0^+} \frac{1/r}{-1/r^2} = \lim_{r\to 0^+} (-r) = 0.$$

So the integrand extends continuously to $[0,\rho]$ and the integral is finite.

Now take $0 < \varepsilon < 1$. For $0 < r \le \varepsilon < 1$ we have $\log r < 0$, so $|\log r| = -\log r$. Integrating by parts with antiderivative $\frac{d}{dr}\left(\frac{r^2}{2}\log r - \frac{r^2}{4}\right) = r\log r + \frac{r}{2} - \frac{r}{2} = r\log r$:

$$\int_0^\varepsilon r|\log r|\,dr = -\int_0^\varepsilon r\log r\,dr = -\left[\frac{r^2}{2}\log r - \frac{r^2}{4}\right]_{r=0}^{r=\varepsilon} = -\frac{\varepsilon^2}{2}\log\varepsilon + \frac{\varepsilon^2}{4},$$

where at the lower limit we used $r^2\log r = r\cdot(r\log r) \to 0$. Since $-\log\varepsilon = |\log\varepsilon|$,

$$\int_{B(0,\varepsilon)} |\Phi|\,dy = \frac{\varepsilon^2}{2}|\log\varepsilon| + \frac{\varepsilon^2}{4}.$$

If $\varepsilon \le e^{-1/2}$ then $|\log\varepsilon| \ge \frac{1}{2}$, so $\frac{\varepsilon^2}{4} \le \frac{\varepsilon^2}{2}|\log\varepsilon|$, and the sum is at most $\varepsilon^2|\log\varepsilon|$. Finally, every compact set lies in some ball $B(0,\rho)$, so $\Phi \in L^1_{\mathrm{loc}}(\mathbb{R}^n)$. □

(d) Integrals of $|\Phi|$ over spheres. On the sphere $\partial B(0,\varepsilon)$ the function Φ is constant, so the integral is (value of $|\Phi|$) × (area $n\alpha(n)\varepsilon^{n-1}$). For $0 < \varepsilon < 1$:

$$\int_{\partial B(0,\varepsilon)} |\Phi|\,dS = \begin{cases} \dfrac{1}{2\pi}|\log\varepsilon|\cdot 2\pi\varepsilon = \varepsilon|\log\varepsilon| & (n=2) \\ \dfrac{1}{n(n-2)\alpha(n)\varepsilon^{n-2}}\cdot n\alpha(n)\varepsilon^{n-1} = \dfrac{\varepsilon}{n-2} & (n \ge 3). \end{cases}$$

(e) Derivatives in x versus derivatives in y. Fix x and put $w(y) = f(x-y)$. By the chain rule,

$$\frac{\partial w}{\partial y_i}(y) = -f_{x_i}(x-y), \qquad \frac{\partial^2 w}{\partial y_i^2}(y) = f_{x_i x_i}(x-y).$$

The two minus signs cancel in the second derivative, so

$$\Delta_y f(x-y) = \Delta_x f(x-y), \qquad |D_y f(x-y)| = |Df(x-y)|.$$

(f) Two limits. As $\varepsilon \to 0^+$,

$$\varepsilon|\log\varepsilon| \to 0 \quad \text{and} \quad \varepsilon^2|\log\varepsilon| = \varepsilon\cdot\varepsilon|\log\varepsilon| \to 0,$$

by the L'Hôpital computation in (c).

2.7.2 The Theorem

Theorem 1 (Solving Poisson's equation). *Define u by* (6)*. Then*

(i) $u \in C^2(\mathbb{R}^n)$

and

(ii) $-\Delta u = f$ *in* $\mathbb{R}^n$.

We consequently see that (6) provides us with a formula for a solution of Poisson's equation $-\Delta u = f$ in $\mathbb{R}^n$.

2.7.3 Proof of Theorem 1

Throughout, fix $R_0 > 0$ such that $f(z) = 0$ whenever $|z| \geq R_0$ (possible because f has compact support).

Step 1: $u \in C^2(\mathbb{R}^n)$. *(1a) Moving the variable x onto f.* By commutativity of convolution (substitute $z = x - y$),

$$u(x) = \int_{\mathbb{R}^n} \Phi(x-y) f(y)\, dy = \int_{\mathbb{R}^n} f(x-y)\, \Phi(y)\, dy.$$

The advantage is that now x appears only in the smooth function f, and not in the singular function Φ. We cannot differentiate Φ under the integral sign ($D^2\Phi$ is not locally integrable), but we *can* differentiate f.

(1b) Differentiating under the integral sign. We apply Lemma 0.15 with $U = \mathbb{R}^n$ (so $U_\rho = \mathbb{R}^n$), with $g = f \in C^2(\mathbb{R}^n)$, which vanishes for $|z| > R_0 = \rho$, and with Φ in the role of the locally integrable function (Lemma 2.1). Part (c) of that lemma with $k = 2$ gives $u \in C^2(\mathbb{R}^n)$ and

$$\frac{\partial u}{\partial x_i}(x) = \int_{\mathbb{R}^n} \Phi(y) \frac{\partial f}{\partial x_i}(x-y)\, dy \quad (i = 1, \ldots, n),$$

$$\frac{\partial^2 u}{\partial x_i \partial x_j}(x) = \int_{\mathbb{R}^n} \Phi(y) \frac{\partial^2 f}{\partial x_i \partial x_j}(x-y)\, dy \quad (i, j = 1, \ldots, n). \tag{8}$$

Let us recall *why* the derivative may be taken inside the integral (this is the content of Lemma 0.15):

- By the mean value theorem, $\frac{f(x+he_i-y)-f(x-y)}{h} = f_{x_i}(x-y+\theta he_i)$ for some $\theta \in (0,1)$, and since f_{x_i} is continuous with compact support, it is uniformly continuous. Hence
$$\frac{f(x+he_i-y)-f(x-y)}{h} \to \frac{\partial f}{\partial x_i}(x-y) \quad \text{uniformly on } \mathbb{R}^n \text{ as } h \to 0$$
(Lemma 0.13).
- For $|h| \le 1$ the integrand vanishes unless $y \in B(0,|x|+R_0+1) =: K$, and $\int_K |\Phi| < \infty$ by Lemma 2.1. So the error is at most $\big(\sup|\text{difference quotient} - f_{x_i}|\big)\int_K |\Phi| \to 0$ (Lemma 0.12).
- The same argument applied to f_{x_i} gives (8); and the right hand side of (8) is continuous in x because $f_{x_ix_j}$ is uniformly continuous. Hence $u \in C^2(\mathbb{R}^n)$, proving (i).

Step 2: Isolating the singularity. Summing (8) over $i = j = 1, \dots, n$ gives
$$\Delta u(x) = \int_{\mathbb{R}^n} \Phi(y)\Delta_x f(x-y)\,dy.$$
Since Φ blows up at 0, we isolate this singularity inside a small ball. Fix ε with $0 < \varepsilon \le e^{-1/2}$ (in particular $\varepsilon < 1$). Then
$$\begin{aligned}\Delta u(x) &= \int_{B(0,\varepsilon)} \Phi(y)\Delta_x f(x-y)\,dy + \int_{\mathbb{R}^n - B(0,\varepsilon)} \Phi(y)\Delta_x f(x-y)\,dy \\ &=: I_\varepsilon + J_\varepsilon.\end{aligned} \tag{9}$$

Estimate of I_ε. For every z, $|\Delta f(z)| = |\sum_{i=1}^n f_{x_ix_i}(z)| \le \sum_{i=1}^n |f_{x_ix_i}(z)| \le n|D^2 f(z)| \le n\|D^2 f\|_{L^\infty(\mathbb{R}^n)}$, because each entry of the Hessian is bounded by its norm $|D^2 f(z)|$. Hence, using Lemma 2.1,
$$|I_\varepsilon| \le n\|D^2 f\|_{L^\infty(\mathbb{R}^n)} \int_{B(0,\varepsilon)} |\Phi(y)|\,dy \le \begin{cases} C\varepsilon^2|\log\varepsilon| & (n=2) \\ C\varepsilon^2 & (n \ge 3), \end{cases} \tag{10}$$
where C is a constant depending only on n and f (namely $C = 2\|D^2 f\|_{L^\infty}$ for $n = 2$ and $C = \frac{n}{2(n-2)}\|D^2 f\|_{L^\infty}$ for $n \ge 3$). In particular $I_\varepsilon \to 0$ as $\varepsilon \to 0$, by (f).

Making J_ε an integral over a bounded region. Green's formulas need a *bounded* region, so we cut off far away. Let $R = |x| + R_0 + 1$. If $|y| \ge R - 1$, then $|x-y| \ge |y| - |x| \ge R_0$, so $f(x-y)$ and all its derivatives vanish there. Set
$$U = \{y : \varepsilon < |y| < R\}.$$
Then $J_\varepsilon = \int_U \Phi(y)\Delta_y f(x-y)\,dy$, where we used (e) to replace Δ_x by Δ_y. (The sphere $|y| = \varepsilon$ has volume zero, so it does not matter whether it is included.) On $\bar{U}$ we have $|y| \ge \varepsilon > 0$, so $\Phi \in C^\infty(\bar{U})$; and $w(y) := f(x-y)$ is in $C^2(\bar{U})$. The boundary of U is smooth, and its outward unit normal ν was described in Example 0.21(b). On the outer sphere $\partial B(0,R)$ we have $w = 0$ and $Dw = 0$.

First integration by parts. Apply Green's formula (ii) (Theorem 0.24) on U with $u = \Phi$ and $v = w$:
$$\int_U Dw \cdot D\Phi\,dy = -\int_U \Phi\Delta w\,dy + \int_{\partial U} \frac{\partial w}{\partial \nu}\Phi\,dS.$$
Rearranging, and dropping the outer sphere where $Dw = 0$,
$$\begin{aligned}J_\varepsilon = \int_U \Phi\Delta w\,dy &= -\int_{\mathbb{R}^n - B(0,\varepsilon)} D\Phi(y)\cdot D_y f(x-y)\,dy \\ &\quad + \int_{\partial B(0,\varepsilon)} \Phi(y)\frac{\partial f}{\partial \nu}(x-y)\,dS(y) \\ &=: K_\varepsilon + L_\varepsilon,\end{aligned} \tag{11}$$

ν denoting the *inward* pointing unit normal along $\partial B(0,\varepsilon)$ (inward for the ball, i.e. outward for U). Here $\frac{\partial f}{\partial \nu}(x-y)$ stands for $\nu \cdot D_y\big(f(x-y)\big)$.

Estimate of L_ε. By the Cauchy–Schwarz inequality and (e), $|\nu \cdot D_y f(x-y)| \le |\nu|\,|Df(x-y)| = |Df(x-y)| \le \|Df\|_{L^\infty(\mathbb{R}^n)}$, since $|\nu| = 1$. Using (d),

$$|L_\varepsilon| \le \|Df\|_{L^\infty(\mathbb{R}^n)} \int_{\partial B(0,\varepsilon)} |\Phi(y)|\, dS(y) \le \begin{cases} C\varepsilon|\log\varepsilon| & (n=2) \\ C\varepsilon & (n \ge 3), \end{cases} \tag{12}$$

with $C = \|Df\|_{L^\infty}$ for $n = 2$ and $C = \frac{1}{n-2}\|Df\|_{L^\infty}$ for $n \ge 3$. In particular $L_\varepsilon \to 0$ as $\varepsilon \to 0$, by (f).

Step 3: The term K_ε carries the value $f(x)$. We integrate by parts once again in the term K_ε. Apply Green's formula (ii) on U, now with $u = w$ and $v = \Phi$:

$$\int_U D\Phi \cdot Dw\, dy = -\int_U w\Delta\Phi\, dy + \int_{\partial U} \frac{\partial \Phi}{\partial \nu} w\, dS.$$

Since $K_\varepsilon = -\int_U D\Phi \cdot Dw\, dy$ and $w = 0$ on the outer sphere,

$$\begin{aligned} K_\varepsilon &= \int_{\mathbb{R}^n - B(0,\varepsilon)} \Delta\Phi(y) f(x-y)\, dy - \int_{\partial B(0,\varepsilon)} \frac{\partial \Phi}{\partial \nu}(y) f(x-y)\, dS(y) \\ &= -\int_{\partial B(0,\varepsilon)} \frac{\partial \Phi}{\partial \nu}(y) f(x-y)\, dS(y), \end{aligned}$$

since Φ is harmonic away from the origin (fact (b)).

Computing the normal derivative. By (7), $D\Phi(y) = \frac{-1}{n\alpha(n)} \frac{y}{|y|^n}$ $(y \ne 0)$, and $\nu = \frac{-y}{|y|} = -\frac{y}{\varepsilon}$ on $\partial B(0,\varepsilon)$. Consequently, for $|y| = \varepsilon$,

$$\frac{\partial \Phi}{\partial \nu}(y) = \nu \cdot D\Phi(y) = \left(-\frac{y}{\varepsilon}\right) \cdot \left(-\frac{1}{n\alpha(n)} \frac{y}{\varepsilon^n}\right) = \frac{y \cdot y}{n\alpha(n)\varepsilon^{n+1}} = \frac{\varepsilon^2}{n\alpha(n)\varepsilon^{n+1}} = \frac{1}{n\alpha(n)\varepsilon^{n-1}}.$$

This is a constant on the sphere, so it comes out of the integral. Since $n\alpha(n)\varepsilon^{n-1}$ is the surface area of the sphere $\partial B(0,\varepsilon)$, we have

$$K_\varepsilon = -\frac{1}{n\alpha(n)\varepsilon^{n-1}} \int_{\partial B(0,\varepsilon)} f(x-y)\, dS(y) = -\fint_{\partial B(x,\varepsilon)} f(y)\, dS(y). \tag{13}$$

In the last equality we substituted $z = x - y$. The map $y \mapsto x - y$ is a reflection followed by a translation, so it preserves distances and surface measure, and it maps $\partial B(0,\varepsilon)$ onto $\partial B(x,\varepsilon)$ (because $|z - x| = |y| = \varepsilon$). The slash through the integral denotes the average (Definition 0.4).

The limit of K_ε. Since the average of the constant 1 is 1, we can write

$$f(x) = \fint_{\partial B(x,\varepsilon)} f(x)\, dS(y).$$

Hence

$$\left| \fint_{\partial B(x,\varepsilon)} f(y)\, dS(y) - f(x) \right| = \left| \fint_{\partial B(x,\varepsilon)} \big(f(y) - f(x)\big)\, dS(y) \right| \le \sup_{|y-x|=\varepsilon} |f(y) - f(x)|.$$

By continuity of f at x, the right hand side tends to 0 as $\varepsilon \to 0$. Therefore

$$K_\varepsilon = -\fint_{\partial B(x,\varepsilon)} f(y)\, dS(y) \to -f(x) \quad \text{as } \varepsilon \to 0.$$

Step 4: Conclusion. Combining (9), (11) and (13), for every $0 < \varepsilon \le e^{-1/2}$,

$$\Delta u(x) = I_\varepsilon + K_\varepsilon + L_\varepsilon.$$

The left side does not depend on ε. Letting $\varepsilon \to 0$ and using (10), (12) and Step 3,

$$I_\varepsilon \to 0, \qquad L_\varepsilon \to 0, \qquad K_\varepsilon \to -f(x).$$

Hence $\Delta u(x) = -f(x)$, that is, $-\Delta u(x) = f(x)$. Since $x \in \mathbb{R}^n$ was arbitrary, this proves (ii). ■

2.8 The Dirac Delta Notation

2.8.1 Motivation: functions of total mass one concentrating at a point

Think of a unit amount of "mass" (or charge, or heat) spread over $\mathbb{R}^n$ with density $\rho(y) \ge 0$, so that $\int_{\mathbb{R}^n} \rho \, dy = 1$. What happens if we squeeze all the mass into smaller and smaller balls around the origin?

A convenient family is given by the mollifiers η_ε (Definition 0.26):

$$\eta_\varepsilon \ge 0, \qquad \int_{\mathbb{R}^n} \eta_\varepsilon \, dy = 1, \qquad \eta_\varepsilon = 0 \text{ outside } B(0,\varepsilon).$$

(Any sequence of nonnegative functions with integral 1 supported in shrinking balls would do; in one dimension one can take the "tents" or the rectangles $\frac{1}{2\varepsilon}\chi_{[-\varepsilon,\varepsilon]}$.)

Pointwise behaviour. For $y \neq 0$, $\eta_\varepsilon(y) = 0$ as soon as $\varepsilon < |y|$, so $\eta_\varepsilon(y) \to 0$. At the origin $\eta_\varepsilon(0) = \varepsilon^{-n}\eta(0) \to \infty$. So, pointwise, the limit "function" would be 0 for $y \neq 0$, $+\infty$ at $y = 0$, and yet it should have total mass 1. *No function does this*: a function which is 0 except at one point has (Lebesgue) integral 0, not 1.

Behaviour against continuous functions. Let g be continuous. Since $\int \eta_\varepsilon = 1$ we can write $g(0) = \int \eta_\varepsilon(y) g(0) \, dy$, and so

$$\begin{aligned}\left| \int_{\mathbb{R}^n} \eta_\varepsilon(y) g(y) \, dy - g(0) \right| &= \left| \int_{B(0,\varepsilon)} \eta_\varepsilon(y) \big(g(y) - g(0)\big) \, dy \right| \\ &\le \sup_{|y| \le \varepsilon} |g(y) - g(0)| \cdot \int \eta_\varepsilon = \sup_{|y| \le \varepsilon} |g(y) - g(0)| \to 0.\end{aligned}$$

Thus

$$\lim_{\varepsilon \to 0} \int_{\mathbb{R}^n} \eta_\varepsilon(y) g(y) \, dy = g(0) \qquad \text{for every continuous } g.$$

So although the functions η_ε have no limit as a function, their action on continuous functions has a perfectly good limit: $g \mapsto g(0)$. This limiting object is called the **Dirac measure** (or Dirac delta) δ_0: the measure on $\mathbb{R}^n$ giving unit mass to the point 0. We use the suggestive notation

$$\int_{\mathbb{R}^n} \delta_0 \, g(y) \, dy := g(0), \qquad \text{and more generally} \qquad \int_{\mathbb{R}^n} \delta_x \, g(y) \, dy := g(x),$$

where δ_x gives unit mass to the point x. (The "integral" here is just notation for "evaluate at the point".) Formally, $\delta_x(y) = \delta_0(x - y)$, the translate of δ_0.

2.8.2 The remark: $-\Delta\Phi = \delta_0$

Remark. We sometimes write

$$-\Delta\Phi = \delta_0 \quad \text{in } \mathbb{R}^n,$$

δ_0 denoting the Dirac measure on $\mathbb{R}^n$ giving unit mass to the point 0. Adopting this notation, we may formally compute:

$$-\Delta u(x) = \int_{\mathbb{R}^n} -\Delta_x \Phi(x-y)\, f(y)\, dy = \int_{\mathbb{R}^n} \delta_x f(y)\, dy = f(x) \quad (x \in \mathbb{R}^n),$$

in accordance with Theorem 1. This corrects the erroneous calculation $\Delta u = \int \Delta_x \Phi(x - y) f(y)\, dy = 0$.

Explanation of the remark.

1. *Why it is reasonable.* $\Delta\Phi(y) = 0$ for every $y \neq 0$ (fact (b) above). So $-\Delta\Phi$ is "zero away from the origin". The only place where something can happen is the singular point $y = 0$, exactly like the limit of the η_ε above.

2. *How much "mass" sits at the origin.* By Green's formula (i), the integral of $-\Delta\Phi$ over a region should equal minus the flux of $D\Phi$ through its boundary. On the sphere $\partial B(0,\varepsilon)$ with *outward* normal $\nu = y/\varepsilon$, (7) gives $\frac{\partial\Phi}{\partial\nu} = -\frac{1}{n\alpha(n)\varepsilon^{n-1}}$, so

$$-\int_{\partial B(0,\varepsilon)} \frac{\partial\Phi}{\partial\nu}\, dS = \frac{1}{n\alpha(n)\varepsilon^{n-1}} \cdot n\alpha(n)\varepsilon^{n-1} = 1 \qquad \text{for every } \varepsilon > 0.$$

   The total "mass" of $-\Delta\Phi$ in every ball around 0 is 1, however small the ball. This is exactly the normalisation we chose when we fixed the constants in Φ.

3. *What the formula really means.* What we actually proved in Theorem 1 is: for $f \in C_c^2(\mathbb{R}^n)$ and every x,

$$\int_{\mathbb{R}^n} \Phi(y)\, \big(-\Delta f\big)(x-y)\, dy = f(x).$$

   Taking $x = 0$ and writing $g(y) = f(-y)$ (so $\Delta g(y) = \Delta f(-y)$; this is (e) with $x = 0$), this says

$$\int_{\mathbb{R}^n} \Phi(y)\, \big(-\Delta g(y)\big)\, dy = g(0) \qquad \text{for all } g \in C_c^2(\mathbb{R}^n).$$

   If we "integrate by parts twice" (moving the Laplacian from g to Φ, as if Φ were smooth), the left side would be $\int(-\Delta\Phi)g\, dy$. So the identity says: $-\Delta\Phi$ *acts on functions* g *exactly as* δ_0 *does*, $g \mapsto g(0)$. This is the precise meaning of $-\Delta\Phi = \delta_0$.

4. *Why the naive calculation fails.* In $\int \Delta_x\Phi(x-y)f(y)\, dy = 0$ we used only that $\Delta_x\Phi(x - y) = 0$ for $y \neq x$, and ignored the single point $y = x$. But all the "mass" of $-\Delta_x\Phi(x - \cdot)$ sits at $y = x$; in the language above, $-\Delta_x\Phi(x-y) = \delta_0(x-y) = \delta_x(y)$. Integrating this against f picks out $f(x)$, not 0.

2.9 Mean-Value Formulas

Consider now an open set $U \subset \mathbb{R}^n$ and suppose u is a harmonic function within U. We next derive the important *mean-value formulas*, which declare that $u(x)$ equals both the average of u over the sphere $\partial B(x,r)$ and the average of u over the entire ball $B(x,r)$, provided $B(x,r) \subset U$.

Theorem 2 (Mean-value formulas for Laplace's equation). *If $u \in C^2(U)$ is harmonic, then*

$$u(x) = \fint_{\partial B(x,r)} u\, dS = \fint_{B(x,r)} u\, dy \tag{14}$$

for each ball $B(x,r) \subset U$.

Remark (the ball must lie inside U). The formula is claimed only for balls $B(x,r)$ which lie *completely* inside U (recall that $B(x,r)$ is the closed ball, so the sphere $\partial B(x,r)$ must also be inside U). This is essential:

- u is only defined (and harmonic) in U; if part of the ball is outside U, the averages do not even make sense.
- The proof uses $\Delta u = 0$ at *every* point of the ball, and $u \in C^2$ up to the sphere (to apply Green's formula on the ball).

By Lemma 0.2, for a point $x \in U$ the admissible radii are exactly $0 < r < \operatorname{dist}(x, \partial U)$ (all $r > 0$ if $U = \mathbb{R}^n$). Every time we use the mean-value formula below, we first check that the ball lies in U.

2.9.1 Proof of Theorem 2

Fix $x \in U$ and let $d = \operatorname{dist}(x, \partial U)$ ($d = \infty$ if $U = \mathbb{R}^n$). For $0 < r < d$, $B(x, r) \subset U$ (Lemma 0.2).

Step 1: the function ϕ. Set

$$\phi(r) := \fint_{\partial B(x,r)} u(y)\, dS(y) = \fint_{\partial B(0,1)} u(x + rz)\, dS(z) \qquad (0 < r < d).$$

The second expression comes from the change of variables $y = x + rz$, which maps the unit sphere onto $\partial B(x, r)$ (Corollary 0.7(b)). The point of this change of variables is that the domain of integration $\partial B(0, 1)$ *no longer depends on* r; now r appears only inside the integrand.

Step 2: differentiating ϕ (interchange of derivative and integral). We claim

$$\phi'(r) = \fint_{\partial B(0,1)} Du(x + rz) \cdot z\, dS(z).$$

Justification. Put $F(r, z) = u(x + rz)$ for $0 < r < d$ and $z \in \partial B(0, 1)$. By the chain rule,

$$\frac{\partial F}{\partial r}(r, z) = \sum_{i=1}^{n} u_{x_i}(x + rz)\, \frac{\partial (x_i + rz_i)}{\partial r} = \sum_{i=1}^{n} u_{x_i}(x + rz)\, z_i = Du(x + rz) \cdot z.$$

Since $u \in C^2(U) \subset C^1(U)$, both F and $\partial F/\partial r$ are continuous on $(0, d) \times \partial B(0, 1)$. The sphere $\partial B(0, 1)$ is compact. Hence Lemma 0.14 allows us to differentiate under the integral sign. (The idea, once more: on a compact set $[a, b] \times \partial B(0, 1)$ the derivative $\partial F/\partial r$ is uniformly continuous, so the difference quotients converge *uniformly* in z, and uniform convergence lets us pass the limit through the integral.) Dividing by the constant $n\alpha(n)$ gives the claim for the average.

Step 3: back to the sphere $\partial B(x, r)$. Undo the change of variables, $z = \frac{y - x}{r}$:

$$\phi'(r) = \fint_{\partial B(x,r)} Du(y) \cdot \frac{y - x}{r}\, dS(y).$$

By Example 0.21(a), $\nu(y) = \frac{y - x}{r}$ is the outward unit normal to the ball $B(x, r)$ at $y \in \partial B(x, r)$. Therefore $Du(y) \cdot \frac{y - x}{r} = \frac{\partial u}{\partial \nu}(y)$ and

$$\phi'(r) = \fint_{\partial B(x,r)} \frac{\partial u}{\partial \nu}\, dS(y).$$

Step 4: Green's formula. Apply Green's formula (i) (Theorem 0.24) on the open ball $B^\circ(x,r)$; this is allowed because $u \in C^2$ on a neighbourhood of the closed ball $B(x,r) \subset U$:

$$\int_{\partial B(x,r)} \frac{\partial u}{\partial \nu}\, dS = \int_{B(x,r)} \Delta u \, dy.$$

Converting both integrals into averages (divide by $n\alpha(n)r^{n-1}$ and use $\int_{B(x,r)} = \alpha(n)r^n \fint_{B(x,r)}$):

$$\phi'(r) = \frac{1}{n\alpha(n)r^{n-1}} \int_{B(x,r)} \Delta u \, dy = \frac{\alpha(n)r^n}{n\alpha(n)r^{n-1}} \fint_{B(x,r)} \Delta u \, dy = \frac{r}{n} \fint_{B(x,r)} \Delta u(y)\, dy.$$

So far we have *not* used that u is harmonic; the identity

$$\phi'(r) = \frac{r}{n} \fint_{B(x,r)} \Delta u(y)\, dy \qquad (0 < r < d) \tag{15}$$

holds for every $u \in C^2(U)$. Now, since $\Delta u = 0$ in U, we get $\phi'(r) = 0$.

Step 5: ϕ is constant, and its value is $u(x)$. Since $\phi' = 0$ on the interval $(0,d)$, ϕ is constant there. To find the constant, let $t \to 0$. Since the average of the constant $u(x)$ is $u(x)$,

$$|\phi(t) - u(x)| = \left| \fint_{\partial B(x,t)} \big(u(y) - u(x)\big)\, dS(y) \right| \le \max_{|y-x|=t} |u(y) - u(x)| \to 0 \quad (t \to 0),$$

by continuity of u at x. Hence, for each $0 < r < d$,

$$\phi(r) = \lim_{t\to 0} \phi(t) = \lim_{t \to 0} \fint_{\partial B(x,t)} u(y)\, dS(y) = u(x).$$

This is the first equality in (14): $\int_{\partial B(x,s)} u \, dS = n\alpha(n)s^{n-1}u(x)$ for $0 < s < d$.

Step 6: the ball average. By polar coordinates centered at x (Corollary 0.7(a)) and Step 5,

$$\int_{B(x,r)} u \, dy = \int_0^r \left(\int_{\partial B(x,s)} u \, dS \right) ds = \int_0^r n\alpha(n)s^{n-1}u(x)\, ds = u(x)\, \alpha(n) r^n.$$

Dividing by $|B(x,r)| = \alpha(n)r^n$ gives $\fint_{B(x,r)} u \, dy = u(x)$. □

2.9.2 Converse to the Mean-Value Property

Theorem 3 (Converse to mean-value property). *If $u \in C^2(U)$ satisfies*

$$u(x) = \fint_{\partial B(x,r)} u \, dS$$

for each ball $B(x,r) \subset U$, then u is harmonic.

Proof. Step 1: setting up a contradiction. Suppose $\Delta u \not\equiv 0$ in U. Then there is a point $x \in U$ with $\Delta u(x) \neq 0$. We may assume $\Delta u(x) > 0$: otherwise replace u by $-u$, which also satisfies the mean-value property (multiply the formula by -1) and has $\Delta(-u)(x) = -\Delta u(x) > 0$.

Step 2: continuity of Δu gives a whole ball where $\Delta u > 0$. Since $u \in C^2(U)$, each $u_{x_i x_i}$ is continuous, so $\Delta u = \sum_i u_{x_i x_i}$ is continuous. With $\eta = \Delta u(x)/2 > 0$ there is $\delta > 0$ such that $|\Delta u(y) - \Delta u(x)| < \eta$, hence $\Delta u(y) > \Delta u(x)/2 > 0$, whenever $|y - x| < \delta$. Choose $0 < r < \min(\delta, \operatorname{dist}(x, \partial U))$. Then $B(x,r) \subset U$ (Lemma 0.2) and $\Delta u > 0$ on $B(x,r)$. *This is where the continuity of Δu is used*: knowing $\Delta u(x) > 0$ at a single point would not be enough, because a single point has no volume.

Step 3: the contradiction. Let $\phi(s) = \fint_{\partial B(x,s)} u\,dS$ as in the proof of Theorem 2. By hypothesis $\phi(s) = u(x)$ for every $0 < s < \operatorname{dist}(x, \partial U)$, so ϕ is constant and $\phi'(r) = 0$. On the other hand, identity (15) (which was derived for *every* $u \in C^2(U)$, without using harmonicity) gives

$$0 = \phi'(r) = \frac{r}{n} \fint_{B(x,r)} \Delta u(y)\,dy > 0,$$

because Δu is continuous and strictly positive on $B(x, r)$, so its integral is positive (Lemma 0.11). This is a contradiction. Hence $\Delta u \equiv 0$ in U. $\square$

2.10 Properties of Harmonic Functions

We now present a sequence of interesting deductions about harmonic functions, all based upon the mean-value formulas. Assume for the following that $U \subset \mathbb{R}^n$ is open and bounded.

2.10.1 Strong Maximum Principle

Theorem 4 (Strong maximum principle). *Suppose $u \in C^2(U) \cap C(\bar{U})$ is harmonic within U.*

(i) *Then*

$$\max_{\bar{U}} u = \max_{\partial U} u.$$

(ii) *Furthermore, if U is connected and there exists a point $x_0 \in U$ such that*

$$u(x_0) = \max_{\bar{U}} u,$$

then u is constant within U.

Assertion (i) is the *maximum principle* for Laplace's equation and (ii) is the *strong maximum principle*.

Why the maxima exist. U is bounded, so $\bar{U}$ is closed and bounded, i.e. compact, and $u \in C(\bar{U})$ attains its maximum on $\bar{U}$. The boundary ∂U is a closed subset of $\bar{U}$, hence compact, and it is nonempty (Lemma 2.2(e) below), so u attains its maximum on ∂U too. Since $\partial U \subset \bar{U}$, we always have $\max_{\partial U} u \le \max_{\bar{U}} u$; the content of (i) is the reverse inequality: *the maximum over the closure is already attained on the boundary.*

The minimum principle. The same statements hold with "min" replacing "max":

(i′) $\min_{\bar{U}} u = \min_{\partial U} u$;

(ii′) if U is connected and there exists $x_0 \in U$ with $u(x_0) = \min_{\bar{U}} u$, then u is constant within U.

Proof from Theorem 4. Let $v = -u$. Then $v \in C^2(U) \cap C(\bar{U})$ and $\Delta v = -\Delta u = 0$, so Theorem 4 applies to v. For any set E on which u attains its minimum,

$$\max_{E} v = \max_{E}(-u) = -\min_{E} u,$$

because $-u(y) \le -u(y_0)$ for all $y \in E$ exactly when $u(y) \ge u(y_0)$ for all $y \in E$: the point where u is smallest is the point where $-u$ is largest.

- (i′): by (i) for v, $-\min_{\bar{U}} u = \max_{\bar{U}} v = \max_{\partial U} v = -\min_{\partial U} u$. Multiply by -1.
- (ii′): if $u(x_0) = \min_{\bar{U}} u$, then $v(x_0) = -\min_{\bar{U}} u = \max_{\bar{U}} v$. By (ii) for v, v is constant in U, hence so is $u = -v$.

Proof of Theorem 4 (ii). Suppose U is connected and there exists a point $x_0 \in U$ with $u(x_0) = M := \max_{\bar{U}} u$.

Step 1: $u \equiv M$ on a ball around x_0. For $0 < r < \operatorname{dist}(x_0, \partial U)$ the ball $B(x_0, r)$ lies in U (Lemma 0.2), so the mean-value property (Theorem 2) applies:

$$M = u(x_0) = \fint_{B(x_0,r)} u \, dy \leq M.$$

The inequality holds because $u \leq M$ everywhere on $\bar{U}$, and the average of a function which is $\leq M$ is $\leq M$. Since the two ends are equal, $\fint_{B(x_0,r)} (M - u) \, dy = M - M = 0$. The function $M - u$ is continuous and ≥ 0, with zero integral over the ball, so by Lemma 0.11, $M - u \equiv 0$, i.e. $u \equiv M$ within $B(x_0, r)$. (As equality holds only if $u \equiv M$ within $B(x_0, r)$, we see $u(y) = M$ for all $y \in B(x_0, r)$.)

Step 2: the set where the maximum is attained. Let

$$A = \{x \in U \mid u(x) = M\}.$$

- $A \neq \emptyset$, since $x_0 \in A$.
- A is open: if $x \in A$, Step 1 applied at x (instead of x_0) shows that a whole ball around x lies in A.
- A is relatively closed in U: $A = U \cap u^{-1}(\{M\})$, and $u^{-1}(\{M\})$ is closed because u is continuous.

Since U is connected, a nonempty subset which is both open and relatively closed must be all of U (Lemma 2.2(a)). Thus $A = U$, i.e. $u \equiv M$ in U. This proves assertion (ii).

Proof of Theorem 4 (i) from (ii). We must show $\max_{\bar{U}} u \leq \max_{\partial U} u$, i.e. that the maximum value $M = \max_{\bar{U}} u$ is taken at some point of ∂U. By compactness of $\bar{U}$ there is a point $x_0 \in \bar{U}$ with $u(x_0) = M$.

Case 1: U connected.

- If $x_0 \in \partial U$, we are done: $M = u(x_0) \leq \max_{\partial U} u$.
- If $x_0 \in U$, then by (ii), $u \equiv M$ in U. Since u is continuous on $\bar{U}$ and every point of ∂U is a limit of points of U, also $u \equiv M$ on $\bar{U}$, in particular on ∂U, which is nonempty. So $\max_{\partial U} u = M$.

Case 2: U not connected. Write U as the disjoint union of its connected components, $U = \bigcup_\lambda U_\lambda$ (Lemma 2.2(c)); each U_λ is open, connected and bounded. Again, if $x_0 \in \partial U$ we are done. Otherwise $x_0 \in U$, so x_0 lies in exactly one component, say U_λ. Now look at u on U_λ only:

- $u \in C^2(U_\lambda) \cap C(\bar{U}_\lambda)$ and $\Delta u = 0$ in U_λ (because $\bar{U}_\lambda \subset \bar{U}$).
- $\max_{\bar{U}_\lambda} u \leq \max_{\bar{U}} u = M = u(x_0)$, and $x_0 \in U_\lambda$; so $u(x_0) = \max_{\bar{U}_\lambda} u$ is attained at an interior point of the connected set U_λ.

By (ii) applied on U_λ, $u \equiv M$ in U_λ, and by continuity $u \equiv M$ on $\bar{U}_\lambda$, in particular on ∂U_λ, which is nonempty since U_λ is bounded (Lemma 2.2(e)). Finally $\partial U_\lambda \subset \partial U$ (Lemma 2.2(d)). Hence there is a point of ∂U where $u = M$, and $\max_{\partial U} u = M$. This proves (i). □

Remark (positivity). The strong maximum principle asserts in particular that if U is connected and $u \in C^2(U) \cap C(\bar{U})$ satisfies

$$\begin{cases} \Delta u = 0 & \text{in } U \\ u = g & \text{on } \partial U, \end{cases}$$

where $g \geq 0$, then u is positive *everywhere* in U if g is positive *somewhere* on ∂U.

Proof of the remark.

1. $u \geq 0$. By the minimum principle (i′), $\min_{\bar{U}} u = \min_{\partial U} u = \min_{\partial U} g \geq 0$. So $u \geq 0$ on $\bar{U}$.

2. *u cannot vanish inside U.* Suppose $u(x_0) = 0$ for some $x_0 \in U$. Since $u \geq 0$, the value 0 is the minimum of u over $\bar{U}$, and it is attained at the *interior* point x_0. As U is connected, the strong minimum principle (ii′) says u is constant in U, so $u \equiv 0$ in U.

3. *Contradiction.* By continuity of u on $\bar{U}$, $u \equiv 0$ on $\bar{U}$, in particular $g = u = 0$ on all of ∂U. This contradicts the assumption that g is positive at some point of ∂U.

Therefore $u(x) > 0$ for every $x \in U$. Note how strong this is: a positive boundary value at a single point (even if $g = 0$ on most of ∂U) makes u positive at every point inside. Connectedness is needed: if U has two components and $g > 0$ only on the boundary of one of them, then $u \equiv 0$ on the other. □

2.10.2 A Topological Remark

The proof of Theorem 4(i) used some facts about connected sets and boundaries, which we now prove.

Lemma 2.2. *Let $U \subset \mathbb{R}^n$ be open.*

(a) (Connectedness.) *U is called connected if it cannot be written as $U = A \cup B$ with A, B disjoint, nonempty and open. Equivalently: if $A \subset U$ is nonempty, open, and relatively closed in U (i.e. $A = U \cap F$ with F closed), then $A = U$.*

(b) If C_λ ($\lambda \in \Lambda$) are connected subsets of U which all contain a common point p, then $\bigcup_\lambda C_\lambda$ is connected. Balls are connected.

(c) (Connected components.) *For $x, y \in U$ write $x \sim y$ if there is a connected set $C \subset U$ with $x, y \in C$. This is an equivalence relation. Each equivalence class $U_x = [x]$ is connected and open. Hence U is the disjoint union of its equivalence classes,*

$$U = \bigcup_\lambda U_\lambda,$$

called the connected components *of U.*

(d) If $U = \bigcup_\lambda U_\lambda$ is a union of pairwise disjoint open sets (for example, its connected components), then $\partial U_\lambda \subset \partial U$ for each λ.

(e) If U is nonempty and bounded, then $\partial U \neq \emptyset$.

Proof. (a) If A is nonempty, open and relatively closed with $A \neq U$, then $B = U \setminus A = U \cap F^c$ is open, nonempty and disjoint from A, and $U = A \cup B$, so U is not connected. Conversely, if $U = A \cup B$ with A, B disjoint, nonempty and open, then $A = U \setminus B = U \cap B^c$ is relatively closed, open, nonempty, and $A \neq U$.

(b) Let $C = \bigcup C_\lambda$ and suppose $C = A \cup B$ with A, B disjoint and open relative to C. Say $p \in A$. Each $C_\lambda = (C_\lambda \cap A) \cup (C_\lambda \cap B)$ is a splitting of the connected set C_λ into disjoint relatively open sets, and $p \in C_\lambda \cap A \neq \emptyset$; so $C_\lambda \cap B = \emptyset$ for every λ, hence $B = \emptyset$. So C is connected. A ball is convex, hence path connected, hence connected (we use this standard fact without proof).

(c) *Equivalence relation.* Reflexive: $\{x\}$ is connected. Symmetric: obvious. Transitive: if $x, y \in C_1$ and $y, z \in C_2$ with C_1, C_2 connected, then $C_1 \cup C_2$ is connected by (b) (common point y) and contains x and z.

Each class is connected. $[x]$ is the union of all connected subsets of U containing x: every such set lies in $[x]$, and every $y \in [x]$ lies in such a set. By (b) (common point x) this union is connected.

Each class is open. Let $y \in [x]$. Since U is open there is a ball $B^\circ(y,\delta) \subset U$; it is connected and contains y. Then $[x] \cup B^\circ(y,\delta)$ is connected by (b) (common point y), lies in U and contains x; hence it is contained in $[x]$. So $B^\circ(y,\delta) \subset [x]$.

Distinct equivalence classes are disjoint and their union is U, which proves (c).

(d) Let $z \in \partial U_\lambda = \bar{U}_\lambda \setminus U_\lambda$ (recall U_λ is open).

- $z \notin U$: otherwise z lies in some U_μ; $\mu \neq \lambda$ because $z \notin U_\lambda$. But U_μ is an open set containing $z \in \bar{U}_\lambda$, so U_μ meets U_λ, contradicting disjointness.
- $z \in \bar{U}$: because $z \in \bar{U}_\lambda$ and $U_\lambda \subset U$.

So $z \in \bar{U} \setminus U = \partial U$.

(e) If $\partial U = \emptyset$ then $\bar{U} = U$, so U is both open and closed in $\mathbb{R}^n$ and nonempty. Since $\mathbb{R}^n$ is connected, (a) gives $U = \mathbb{R}^n$, which is not bounded. □

Boundaries of unions in general. What is the relation between $\partial(A \cup B)$ and ∂A, ∂B for arbitrary sets?

- *One inclusion always holds:* $\partial(A \cup B) \subset \partial A \cup \partial B$. Indeed, if $x \notin \partial A$ and $x \notin \partial B$, there are neighbourhoods N_A, N_B of x with $N_A \subset A$ or $N_A \subset A^c$, and $N_B \subset B$ or $N_B \subset B^c$. If $N_A \subset A$ or $N_B \subset B$, this neighbourhood lies in $A \cup B$; otherwise $N_A \cap N_B \subset (A \cup B)^c$. Either way $x \notin \partial(A \cup B)$.
- *The reverse inclusion fails in general.* $A = [0,1]$, $B = [1,2]$: $\partial A \cup \partial B = \{0,1,2\}$, but $\partial(A \cup B) = \{0,2\}$. Overlapping open sets: $A = (0,1)$, $B = (\frac{1}{2},2)$: $\partial A \cup \partial B = \{0, \frac{1}{2}, 1, 2\}$ but $\partial(A \cup B) = \{0,2\}$. Extreme case: $A = \mathbb{Q}$, $B = \mathbb{R} \setminus \mathbb{Q}$ have $\partial A = \partial B = \mathbb{R}$, while $\partial(A \cup B) = \partial\mathbb{R} = \emptyset$.
- *Infinite unions:* even the first inclusion can fail. $A_k = [\frac{1}{k}, 1]$ has $\partial A_k = \{\frac{1}{k}, 1\}$, but $\bigcup_k A_k = (0,1]$ has boundary $\{0,1\}$, and $0 \notin \bigcup_k \partial A_k$.
- *For pairwise disjoint open sets* (as in (d)) we get equality, $\partial U = \bigcup_\lambda \partial U_\lambda$, when there are finitely many sets: "$\supset$" is (d) and "$\subset$" is the first bullet. The proof of (d) does not need finiteness, which is why it applies to components even if there are infinitely many of them.

2.10.3 Uniqueness

An important application of the maximum principle is establishing the uniqueness of solutions to certain boundary-value problems for Poisson's equation.

Theorem 5 (Uniqueness). *Let $g \in C(\partial U)$, $f \in C(U)$. Then there exists at most one solution $u \in C^2(U) \cap C(\bar{U})$ of the boundary-value problem*

$$\begin{cases} -\Delta u = f & \text{in } U \\ u = g & \text{on } \partial U. \end{cases} \tag{16}$$

Proof. Suppose u and $\tilde{u}$ both satisfy (16), with $u, \tilde{u} \in C^2(U) \cap C(\bar{U})$.

Step 1: the difference is harmonic with zero boundary values. Let $w := u - \tilde{u}$. Then $w \in C^2(U) \cap C(\bar{U})$ and, because the Laplacian is linear,

$$\Delta w = \Delta u - \Delta \tilde{u} = (-f) - (-f) = 0 \text{ in } U, \qquad w = g - g = 0 \text{ on } \partial U.$$

The unknown right hand side f has disappeared; this is why we can use the maximum principle, which is a statement about *harmonic* functions.

Step 2: $w \leq 0$. By the maximum principle (Theorem 4(i)) applied to w,

$$\max_{\bar{U}} w = \max_{\partial U} w = 0, \qquad \text{so } w \leq 0 \text{ on } \bar{U}.$$

Step 3: $w \geq 0$. The function $-w = \tilde{u} - u$ is also harmonic in U, in $C^2(U) \cap C(\bar{U})$, and zero on ∂U. By Theorem 4(i) applied to $-w$, $\max_{\bar{U}}(-w) = 0$, so $-w \leq 0$, i.e. $w \geq 0$. (Equivalently, use the minimum principle (i′) for w.)

Step 4: conclusion. From Steps 2 and 3, $w \equiv 0$ on $\bar{U}$, that is, $u \equiv \tilde{u}$. □

Remark. Boundedness of U is essential (it was used to have the maximum principle). In the unbounded half-space $U = \{x \in \mathbb{R}^n : x_n > 0\}$, both $u \equiv 0$ and $u(x) = x_n$ are harmonic, continuous up to $\partial U = \{x_n = 0\}$, and vanish on ∂U. So uniqueness fails there.

2.10.4 Regularity

Now we prove that if $u \in C^2$ is harmonic, then necessarily $u \in C^\infty$. Thus *harmonic functions are automatically infinitely differentiable.* This sort of assertion is called a *regularity* theorem. The interesting point is that the algebraic structure of Laplace's equation $\Delta u = \sum_{i=1}^n u_{x_i x_i} = 0$ leads to the analytic deduction that all the partial derivatives of u exist, even those which do not appear in the PDE.

Remark (what is surprising here). Being harmonic only tells us that *one particular combination* of second derivatives, namely the sum $u_{x_1x_1} + \cdots + u_{x_nx_n}$, is zero. It says nothing directly about $u_{x_1x_2}$, or about any third derivative. Yet the conclusion is that *all* partial derivatives of *all* orders exist and are continuous, i.e. $u \in C^\infty(U)$. The route is through the mean-value property: Theorem 2 converts the PDE into an averaging identity, and averaging against a smooth kernel produces smooth functions.

Theorem 6 (Smoothness). *If $u \in C(U)$ satisfies the mean-value property* (14) *for each ball $B(x, r) \subset U$, then*

$$u \in C^\infty(U).$$

Note carefully that u may not be smooth, or even continuous, up to ∂U. The theorem is an *interior* statement.

Corollary. If $u \in C^2(U)$ is harmonic, then u satisfies the mean-value property (Theorem 2), so $u \in C^\infty(U)$ by Theorem 6. Since u is C^∞, partial derivatives commute and for any multiindex β, $\Delta(D^\beta u) = D^\beta(\Delta u) = 0$: *every partial derivative of a harmonic function is again harmonic.*

Remark (notation for radial functions). A function f on $\mathbb{R}^n$ is radial if $f(x)$ depends only on $|x|$: $f(x) = \tilde{f}(|x|)$ for a function $\tilde{f}$ of one variable. It is customary to use *the same letter* for both functions and write $f(x) = f(|x|)$, or $f(r)$ with $r = |x|$. We did this for Φ (where $\Phi(x) = v(|x|)$), and we do it now for the mollifier: $\eta(x) = \eta(|x|)$, where on the right $\eta(r) = C \exp\left(\frac{1}{r^2-1}\right)$ for $0 \leq r < 1$ and $\eta(r) = 0$ for $r \geq 1$. Thus

$$\eta_\varepsilon(x - y) = \frac{1}{\varepsilon^n}\eta\left(\frac{x-y}{\varepsilon}\right) = \frac{1}{\varepsilon^n}\eta\left(\frac{|x-y|}{\varepsilon}\right).$$

Idea of the proof. Let $u^\varepsilon = \eta_\varepsilon * u$ be the mollification of u. From Section 0 we know $u^\varepsilon \in C^\infty(U_\varepsilon)$ (Theorem 0.28(i)). We will show that the mean-value property forces

$$u^\varepsilon = u \quad \text{on } U_\varepsilon,$$

so u itself is C^∞ on U_ε. Being differentiable is a *local* property: to know all derivatives of u at a point x, it is enough to know u on some open set containing x. Given $x \in U$, we choose $\varepsilon = \frac{1}{2}\operatorname{dist}(x, \partial U)$; then $x \in U_\varepsilon$, and u is smooth near x.

Proof of Theorem 6. Let η be a standard mollifier (Definition 0.26), and recall that η is a radial function. Set $u^\varepsilon := \eta_\varepsilon * u$ in $U_\varepsilon = \{x \in U \mid \operatorname{dist}(x, \partial U) > \varepsilon\}$.

Step 1: u^ε is defined and smooth on U_ε. u is continuous on U, hence locally integrable (Example 0.9(3)). By Theorem 0.28(i), $u^\varepsilon \in C^\infty(U_\varepsilon)$.

Step 2: $u^\varepsilon = u$ on U_ε. We will prove u is smooth by demonstrating that in fact $u \equiv u^\varepsilon$ on U_ε. Fix $x \in U_\varepsilon$. Since $\operatorname{dist}(x, \partial U) > \varepsilon$, the closed ball $B(x, \varepsilon)$ lies in U, and so does every $B(x, r)$, $0 < r \le \varepsilon$ (Lemma 0.2). Then:

$$\begin{aligned}
u^\varepsilon(x) &= \int_U \eta_\varepsilon(x-y)u(y)\,dy && \text{(definition)}\\
&= \frac{1}{\varepsilon^n}\int_{B(x,\varepsilon)} \eta\left(\frac{|x-y|}{\varepsilon}\right)u(y)\,dy && (\eta_\varepsilon(x-y)=0 \text{ if } |x-y|\ge\varepsilon\text{; radial notation})\\
&= \frac{1}{\varepsilon^n}\int_0^\varepsilon \eta\left(\frac{r}{\varepsilon}\right)\left(\int_{\partial B(x,r)} u\,dS\right)dr && \text{(polar coordinates about } x)\\
&= \frac{1}{\varepsilon^n}u(x)\int_0^\varepsilon \eta\left(\frac{r}{\varepsilon}\right)n\alpha(n)r^{n-1}\,dr && \text{(mean-value property)}\\
&= u(x)\int_{B(0,\varepsilon)} \eta_\varepsilon\,dy = u(x).
\end{aligned}$$

Let us explain each line.

(1) *Restricting to the ball.* $\eta_\varepsilon(z) = 0$ for $|z| \ge \varepsilon$, so only $y \in B(x, \varepsilon)$ contribute; and $\eta_\varepsilon(x-y) = \varepsilon^{-n}\eta(|x-y|/\varepsilon)$ by the radial notation.

(2) *Polar coordinates.* We use Corollary 0.7(a) on the ball $B(x, \varepsilon)$ centered at x, for the continuous function $y \mapsto \varepsilon^{-n}\eta(|x-y|/\varepsilon)u(y)$. On the sphere $\partial B(x, r)$ we have $|x-y| = r$, so the factor $\eta(r/\varepsilon)$ is *constant on each sphere* and comes out of the inner integral.

(3) *Mean-value property.* Since $B(x, r) \subset U$ for $0 < r \le \varepsilon$, the hypothesis (14) gives $⨍_{\partial B(x,r)} u\,dS = u(x)$, i.e. $\int_{\partial B(x,r)} u\,dS = n\alpha(n)r^{n-1}u(x)$. The constant $u(x)$ comes out of the r-integral.

(4) *Polar coordinates backwards.* Apply Corollary 0.7(c) to the radial function $\eta_\varepsilon(y) = \varepsilon^{-n}\eta(|y|/\varepsilon)$ on $B(0, \varepsilon)$: $\int_{B(0,\varepsilon)} \eta_\varepsilon\,dy = n\alpha(n)\int_0^\varepsilon \varepsilon^{-n}\eta(r/\varepsilon)r^{n-1}\,dr$. And $\int_{B(0,\varepsilon)} \eta_\varepsilon = \int_{\mathbb{R}^n} \eta_\varepsilon = 1$.

Thus $u^\varepsilon \equiv u$ in U_ε, and so $u \in C^\infty(U_\varepsilon)$ for each $\varepsilon > 0$.

Step 3: $u \in C^\infty(U)$. Let $x \in U$ and put $\varepsilon = \frac{1}{2}\operatorname{dist}(x, \partial U) > 0$ (if $U = \mathbb{R}^n$, any $\varepsilon > 0$). Then $\operatorname{dist}(x, \partial U) = 2\varepsilon > \varepsilon$, so $x \in U_\varepsilon$. The set U_ε is open (Lemma 0.2(c)), and on it $u = u^\varepsilon$, which is C^∞. Partial derivatives at x only depend on the values of the function in a neighbourhood of x, so all partial derivatives of u of all orders exist at x and coincide with those of u^ε; they are continuous near x. Since $x \in U$ was arbitrary, $u \in C^\infty(U)$. □

2.10.5 Local Estimates for Harmonic Functions

Next we employ the mean-value formulas to derive careful estimates on the various partial derivatives of a harmonic function. The precise structure of these estimates will be needed below, when we prove analyticity.

Theorem 7 (Estimates on derivatives). *Assume u is harmonic in U. Then*

$$|D^\alpha u(x_0)| \le \frac{C_k}{r^{n+k}}\|u\|_{L^1(B(x_0,r))} \tag{17}$$

for each ball $B(x_0, r) \subset U$ and each multiindex α of order $|\alpha| = k$. Here

$$C_0 = \frac{1}{\alpha(n)}, \qquad C_k = \frac{(2^{n+1}nk)^k}{\alpha(n)} \quad (k = 1, \dots). \tag{18}$$

Recall $\|u\|_{L^1(B(x_0,r))} = \int_{B(x_0,r)} |u|\,dy$ and $\|u\|_{L^\infty(E)} = \sup_E |u|$. By the Corollary to Theorem 6, $u \in C^\infty(U)$, so all $D^\alpha u$ exist, and each $D^\beta u$ is harmonic.

Proof. We establish (17), (18) by induction on k.

The case $k = 0$. Here $\alpha = 0$ and $D^\alpha u = u$. Let $B(x_0, r) \subset U$. The mean-value formula (Theorem 2) gives the *equality*

$$u(x_0) = \fint_{B(x_0,r)} u\,dy = \frac{1}{\alpha(n)r^n}\int_{B(x_0,r)} u\,dy,$$

and taking absolute values (the absolute value of an integral is at most the integral of the absolute value),

$$|u(x_0)| \le \frac{1}{\alpha(n)r^n}\int_{B(x_0,r)} |u|\,dy = \frac{C_0}{r^n}\|u\|_{L^1(B(x_0,r))}.$$

This is (17) with $k = 0$, $C_0 = 1/\alpha(n)$.

The case $k = 1$. Now $|\alpha| = 1$, so $D^\alpha u = u_{x_i}$ for some i. Let $B(x_0, r) \subset U$. The proof has two steps.

Step 1: estimate $u_{x_i}(x_0)$ by the values of u on the sphere $\partial B(x_0, r/2)$. Differentiating Laplace's equation, u_{x_i} $(i = 1, \dots, n)$ is harmonic (Corollary to Theorem 6). The ball $B(x_0, r/2)$ lies in $B(x_0, r) \subset U$, so the mean-value formula applies to u_{x_i} on it:

$$\begin{aligned} |u_{x_i}(x_0)| &= \left|\fint_{B(x_0,r/2)} u_{x_i}\,dx\right| = \frac{2^n}{\alpha(n)r^n}\left|\int_{B(x_0,r/2)} u_{x_i}\,dx\right| \\ &= \frac{2^n}{\alpha(n)r^n}\left|\int_{\partial B(x_0,r/2)} u\nu_i\,dS\right| \\ &\le \frac{2^n}{\alpha(n)r^n}\,\|u\|_{L^\infty(\partial B(x_0,\frac{r}{2}))}\,n\alpha(n)\left(\frac{r}{2}\right)^{n-1} = \frac{2n}{r}\|u\|_{L^\infty(\partial B(x_0,\frac{r}{2}))}. \end{aligned} \tag{19}$$

The details:

- $|B(x_0, r/2)| = \alpha(n)(r/2)^n = \alpha(n)r^n/2^n$, which gives the factor $2^n/(\alpha(n)r^n)$.
- The second line is the Gauss–Green theorem (Theorem 0.22) on the ball $B^\circ(x_0, r/2)$, with outward normal $\nu = (\nu_1, \dots, \nu_n)$; it applies since u is C^1 on a neighbourhood of the closed ball.
- For the inequality: $|u\nu_i| \le |u|\,|\nu| = |u| \le \|u\|_{L^\infty(\partial B(x_0,r/2))}$ on the sphere (the components of a unit vector are at most 1 in absolute value), and the area of the sphere is $n\alpha(n)(r/2)^{n-1}$.
- Simplifying the constant: $\frac{2^n}{\alpha(n)r^n}\cdot n\alpha(n)\frac{r^{n-1}}{2^{n-1}} = \frac{2n}{r}$.

Step 2: estimate u on the sphere $\partial B(x_0, r/2)$ by $\|u\|_{L^1(B(x_0,r))}$. Now if $x \in \partial B(x_0, r/2)$, then $B(x, r/2) \subset B(x_0, r) \subset U$: for $|z - x| \le r/2$, $|z - x_0| \le |z - x| + |x - x_0| \le \frac{r}{2} + \frac{r}{2} = r$. So we may apply the case $k = 0$ at the point x with radius $r/2$:

$$|u(x)| \le \frac{1}{\alpha(n)}\left(\frac{2}{r}\right)^n \|u\|_{L^1(B(x,r/2))} \le \frac{1}{\alpha(n)}\left(\frac{2}{r}\right)^n \|u\|_{L^1(B(x_0,r))}.$$

In the last step we enlarged the domain of integration of $|u| \ge 0$ from $B(x, r/2)$ to $B(x_0, r)$. The right hand side *does not depend on the point* x of the sphere. Taking the supremum over $x \in \partial B(x_0, r/2)$,

$$\|u\|_{L^\infty(\partial B(x_0,\frac{r}{2}))} \le \frac{1}{\alpha(n)}\left(\frac{2}{r}\right)^n \|u\|_{L^1(B(x_0,r))}.$$

Combining with Step 1, we deduce

$$|D^\alpha u(x_0)| \le \frac{2n}{r}\cdot\frac{2^n}{\alpha(n)r^n}\|u\|_{L^1(B(x_0,r))} = \frac{2^{n+1}n}{\alpha(n)}\frac{1}{r^{n+1}}\|u\|_{L^1(B(x_0,r))}$$

if $|\alpha| = 1$. This verifies (17), (18) for $k = 1$, since $C_1 = \frac{2^{n+1}n}{\alpha(n)}$.

The induction step. Assume now $k \geq 2$ and that (17), (18) are valid for all balls in U and each multiindex of order less than or equal to $k-1$. Explicitly, the *induction hypothesis* we use is: for every ball $B(x,\rho) \subset U$ and every multiindex β with $|\beta| = k-1$,

$$|D^\beta u(x)| \leq \frac{(2^{n+1}n(k-1))^{k-1}}{\alpha(n)\,\rho^{n+k-1}} \|u\|_{L^1(B(x,\rho))}.$$

Fix $B(x_0,r) \subset U$ and let α be a multiindex with $|\alpha| = k$. Choose i with $\alpha_i \geq 1$ and let $\beta = \alpha - e_i$ (lower the i-th entry by one). Then $|\beta| = k-1$ and $D^\alpha u = (D^\beta u)_{x_i}$. Write

$$w := D^\beta u, \qquad \text{so that} \qquad D^\alpha u = w_{x_i}.$$

Now we simply *imitate the case* $k = 1$, with w in place of u: there we estimated u_{x_i}, now we estimate w_{x_i}. The only change is the choice of radii: we use the ball $B(x_0, r/k)$ instead of $B(x_0, r/2)$, and balls of radius $r - \frac{r}{k} = \frac{k-1}{k}r$ instead of $\frac{r}{2}$.

Step 1 (as in (19)*).* $w = D^\beta u$ is harmonic, hence so is w_{x_i}, and $B(x_0, r/k) \subset B(x_0, r) \subset U$. By the mean-value formula for w_{x_i} on $B(x_0, r/k)$ and the Gauss–Green theorem on that ball,

$$\begin{aligned} |w_{x_i}(x_0)| &= \frac{k^n}{\alpha(n) r^n} \left| \int_{B(x_0, r/k)} w_{x_i}\, dx \right| = \frac{k^n}{\alpha(n) r^n} \left| \int_{\partial B(x_0, r/k)} w \nu_i \, dS \right| \\ &\leq \frac{k^n}{\alpha(n) r^n}\, n\alpha(n) \left(\frac{r}{k}\right)^{n-1} \|w\|_{L^\infty(\partial B(x_0, \frac{r}{k}))}. \end{aligned}$$

Since $\frac{k^n}{r^n} \cdot n \frac{r^{n-1}}{k^{n-1}} = \frac{nk}{r}$, this is

$$|D^\alpha u(x_0)| \leq \frac{nk}{r} \|D^\beta u\|_{L^\infty(\partial B(x_0, \frac{r}{k}))}.$$

Step 2 (bound $D^\beta u$ *on the sphere* $\partial B(x_0, r/k)$*).* If $x \in \partial B(x_0, \frac{r}{k})$, then $B(x, \frac{k-1}{k}r) \subset B(x_0, r) \subset U$: for $|z - x| \leq \frac{k-1}{k}r$,

$$|z - x_0| \leq |z - x| + |x - x_0| \leq \frac{k-1}{k}r + \frac{r}{k} = r.$$

So the induction hypothesis applies at x with $\rho = \frac{k-1}{k}r$:

$$|D^\beta u(x)| \leq \frac{(2^{n+1}n(k-1))^{k-1}}{\alpha(n)\left(\frac{k-1}{k}r\right)^{n+k-1}} \|u\|_{L^1(B(x, \frac{k-1}{k}r))} \leq \frac{(2^{n+1}n(k-1))^{k-1}}{\alpha(n)\left(\frac{k-1}{k}r\right)^{n+k-1}} \|u\|_{L^1(B(x_0,r))}.$$

Again the right side does not depend on x, so it bounds $\|D^\beta u\|_{L^\infty(\partial B(x_0, \frac{r}{k}))}$.

Combining the two steps. We get $|D^\alpha u(x_0)| \leq \frac{Q}{\alpha(n) r^{n+k}} \|u\|_{L^1(B(x_0,r))}$ with

$$Q = nk\,(2^{n+1}n)^{k-1}(k-1)^{k-1} \left(\frac{k}{k-1}\right)^{n+k-1}.$$

Let us simplify. Since $(k-1)^{k-1}\left(\frac{k}{k-1}\right)^{k-1} = k^{k-1}$,

$$Q = nk\,(2^{n+1}n)^{k-1} k^{k-1} \left(\frac{k}{k-1}\right)^{n} = (2^{n+1}n)^{k-1}\, n\, k^k \left(\frac{k}{k-1}\right)^{n}.$$

Because $k \geq 2$ we have $\frac{k}{k-1} \leq 2$, so $\left(\frac{k}{k-1}\right)^n \leq 2^n \leq 2^{n+1}$. Hence

$$Q \leq (2^{n+1}n)^{k-1} \cdot 2^{n+1} n \cdot k^k = (2^{n+1}n)^k k^k = (2^{n+1}nk)^k.$$

Therefore

$$|D^\alpha u(x_0)| \le \frac{(2^{n+1}nk)^k}{\alpha(n)r^{n+k}}\|u\|_{L^1(B(x_0,r))}. \tag{20}$$

This confirms (17), (18) for $|\alpha| = k$. □

Why the radii r/k and $r - r/k$? We need two balls: a small ball $B(x_0, \rho_1)$ for the Gauss–Green step, and balls $B(x, \rho_2)$ centered on its boundary sphere for the induction hypothesis. They must satisfy $\rho_1 + \rho_2 \le r$ to stay inside $B(x_0, r)$. Step 1 costs a factor n/ρ_1 and Step 2 a factor $\rho_2^{-(n+k-1)}$. Choosing $\rho_1 = r/k$ and $\rho_2 = r(1 - 1/k)$ balances these so that the constant comes out exactly in the form $(2^{n+1}nk)^k$. For $k = 1$ this choice would give $\rho_2 = 0$, which is why the case $k = 1$ is done separately with $\rho_1 = \rho_2 = r/2$.

2.10.6 Liouville's Theorem

Next we see that there are no nontrivial bounded harmonic functions on all of $\mathbb{R}^n$.

Theorem 8 (Liouville's Theorem). *Suppose $u : \mathbb{R}^n \to \mathbb{R}$ is harmonic and bounded. Then u is constant.*

Proof. Step 1: the estimate. Let $M = \|u\|_{L^\infty(\mathbb{R}^n)} = \sup_{\mathbb{R}^n} |u| < \infty$. Fix $x_0 \in \mathbb{R}^n$ and $r > 0$. Since $U = \mathbb{R}^n$, every ball $B(x_0, r)$ lies in U, *no matter how large r is.* (Theorem 7 holds for any open set U: its proof only used balls inside U, never the boundedness of U.) Apply Theorem 7 with $k = 1$ on $B(x_0, r)$:

$$|u_{x_i}(x_0)| \le \frac{C_1}{r^{n+1}}\|u\|_{L^1(B(x_0,r))}.$$

Step 2: bound the L^1 norm by the sup norm. $\|u\|_{L^1(B(x_0,r))} = \int_{B(x_0,r)} |u|\,dy \le M\,|B(x_0, r)| = M\alpha(n)r^n$. Hence

$$|u_{x_i}(x_0)| \le \frac{C_1}{r^{n+1}}\,\alpha(n)r^n M = \frac{C_1\alpha(n)}{r}\|u\|_{L^\infty(\mathbb{R}^n)} \to 0 \quad \text{as } r \to \infty.$$

Step 3: $Du \equiv 0$. The left side does not depend on r, so $u_{x_i}(x_0) = 0$, for each i and each x_0. Thus $Du \equiv 0$.

Step 4: u is constant. For any $x \in \mathbb{R}^n$, apply the fundamental theorem of calculus along the segment from 0 to x:

$$u(x) - u(0) = \int_0^1 \frac{d}{dt}u(tx)\,dt = \int_0^1 Du(tx) \cdot x\,dt = 0.$$

So $u(x) = u(0)$ for all x, and u is constant. □

2.10.7 Representation Formula

Theorem 9 (Representation formula). *Let $f \in C_c^2(\mathbb{R}^n)$, $n \ge 3$. Then any bounded solution of*

$$-\Delta u = f \quad \text{in } \mathbb{R}^n$$

has the form

$$u(x) = \int_{\mathbb{R}^n} \Phi(x - y)f(y)\,dy + C \quad (x \in \mathbb{R}^n)$$

for some constant C.

Remark. The theorem is stated only for $n \ge 3$; the proof below does not work for $n = 2$. The reason (explained after the proof) is that for $n \ge 3$, Φ is *bounded away from the origin* (in fact $\Phi(x) \to 0$ as $|x| \to \infty$), whereas for $n = 2$, $\Phi(x) = -\frac{1}{2\pi}\log|x|$ is unbounded as $|x| \to \infty$.

Proof. By a solution we mean $u \in C^2(\mathbb{R}^n)$ with $-\Delta u = f$. Let

$$\tilde{u}(x) := \int_{\mathbb{R}^n} \Phi(x - y)f(y)\,dy.$$

Step 1: $\tilde{u}$ is a solution. By Theorem 1, $\tilde{u} \in C^2(\mathbb{R}^n)$ and $-\Delta\tilde{u} = f$ in $\mathbb{R}^n$.

Step 2: $\tilde{u}$ is bounded (here $n \geq 3$ is used). Choose R_0 with $f = 0$ outside $B(0, R_0)$, and let $M_f = \max|f|$. By commutativity of convolution, $\tilde{u}(x) = \int_{\mathbb{R}^n} \Phi(y) f(x-y)\, dy$. Split the integral into the unit ball and its complement:

$$\tilde{u}(x) = \underbrace{\int_{B(0,1)} \Phi(y) f(x-y)\, dy}_{=:A(x)} + \underbrace{\int_{\mathbb{R}^n \setminus B(0,1)} \Phi(y) f(x-y)\, dy}_{=:B(x)}.$$

- *On the unit ball* the singularity of Φ is integrable: $\Phi(y) = \frac{1}{n(n-2)\alpha(n)}|y|^{-(n-2)}$ and $n-2 < n$, so by Lemma 2.1 (or Lemma 0.10), $\int_{B(0,1)} |\Phi|\, dy = \frac{1}{2(n-2)}$. Since $|f(x-y)| \leq M_f$,

$$|A(x)| \leq M_f \int_{B(0,1)} |\Phi(y)|\, dy = \frac{M_f}{2(n-2)}.$$

- *Outside the unit ball* Φ is bounded: for $|y| \geq 1$, $|y|^{n-2} \geq 1$ (as $n-2 > 0$), so

$$0 < \Phi(y) = \frac{1}{n(n-2)\alpha(n)|y|^{n-2}} \leq \frac{1}{n(n-2)\alpha(n)} = \Phi(1),$$

  writing $\Phi(1)$ for the value of Φ on the unit sphere. Using the compact support of f (the substitution $z = x - y$),

$$|B(x)| \leq \Phi(1) \int_{\mathbb{R}^n} |f(x-y)|\, dy = \Phi(1) \int_{\mathbb{R}^n} |f(z)|\, dz = \Phi(1) \int_{B(0,R_0)} |f| \leq \Phi(1) M_f\, \alpha(n) R_0^n.$$

Both bounds are independent of x. Hence $|\tilde{u}(x)| \leq \frac{M_f}{2(n-2)} + \Phi(1) M_f \alpha(n) R_0^n$ for all x: $\tilde{u}$ is a bounded solution of $-\Delta u = f$.

Step 3: Liouville. If u is another bounded solution, let $w := u - \tilde{u}$. Then $w \in C^2(\mathbb{R}^n)$,

$$\Delta w = \Delta u - \Delta\tilde{u} = -f + f = 0,$$

so w is harmonic in $\mathbb{R}^n$; and w is bounded, since $|w| \leq |u| + |\tilde{u}|$ and both are bounded. By Liouville's Theorem (Theorem 8), w is constant, say $w \equiv C$. Thus $u = \tilde{u} + C$, which is the claimed formula. □

Remark (why the theorem is not true for $n = 2$). If $n = 2$, $\Phi(x) = -\frac{1}{2\pi}\log|x|$ is unbounded as $|x| \to \infty$, and so may be $\int_{\mathbb{R}^2} \Phi(x-y) f(y)\, dy$. Then Step 2 fails: outside the unit ball we no longer have a bound $|\Phi(y)| \leq \Phi(1)$, and $\tilde{u}$ need not be bounded, so Liouville's theorem cannot be applied to $u - \tilde{u}$. In fact, when $\int f \neq 0$, $\tilde{u}$ is *always* unbounded:

Let $m = \int_{\mathbb{R}^2} f\, dy \neq 0$ and $f = 0$ outside $B(0, R_0)$. For $|x| \geq 2R_0$ and $|y| \leq R_0$ write $\log|x-y| = \log|x| + \log\frac{|x-y|}{|x|}$. Then

$$\tilde{u}(x) = -\frac{1}{2\pi}\int_{B(0,R_0)} \log|x-y|\, f(y)\, dy = -\frac{m}{2\pi}\log|x| - \frac{1}{2\pi}\int_{B(0,R_0)} \log\frac{|x-y|}{|x|}\, f(y)\, dy.$$

By the triangle inequality, $\frac{1}{2} \leq 1 - \frac{R_0}{|x|} \leq \frac{|x-y|}{|x|} \leq 1 + \frac{R_0}{|x|} \leq \frac{3}{2}$, so $\left|\log\frac{|x-y|}{|x|}\right| \leq \log 2$, and the last integral is bounded by $\frac{\log 2}{2\pi}\int|f|$, independently of x. Therefore

$$\left|\tilde{u}(x) + \frac{m}{2\pi}\log|x|\right| \leq \frac{\log 2}{2\pi}\int|f| \quad (|x| \geq 2R_0),$$

and since $\log|x| \to \infty$, $|\tilde{u}(x)| \to \infty$ as $|x| \to \infty$. So for $n = 2$ the functions $\tilde{u} + C$ are *never* bounded, and the conclusion "bounded solutions are of the form $\tilde{u} + C$" cannot be obtained in this way. (If $\int f = 0$ the $\log|x|$ term disappears, $\tilde{u}$ is bounded, and the proof above goes through; but this is a special situation.)

2.10.8 Analyticity

Next we refine Theorem 6. Recall that u is *analytic* in U if near each point $x_0 \in U$ it can be represented by a convergent power series $\sum_\alpha \frac{D^\alpha u(x_0)}{\alpha!}(x-x_0)^\alpha$.

Theorem 10 (Analyticity). *Assume u is harmonic in U. Then u is analytic in U.*

(We omit the proof for now. It uses the precise form of the constants C_k in Theorem 7.)

2.10.9 Harnack's Inequality

Recall that we write $V \subset\subset U$ to mean $V \subset \bar{V} \subset U$ and $\bar{V}$ is compact (Definition 0.1).

Theorem 11 (Harnack's inequality). *For each connected open set $V \subset\subset U$, there exists a positive constant C, depending only on V, such that*

$$\sup_V u \le C \inf_V u$$

for all nonnegative harmonic functions u in U.

Thus in particular

$$\frac{1}{C}u(y) \le u(x) \le Cu(y)$$

for all points $x, y \in V$. These inequalities assert that *the values of a nonnegative harmonic function within V are all comparable*: u cannot be very small (or very large) at any point of V unless u is very small (or very large) everywhere in V. The intuitive idea is that since V is a positive distance away from ∂U, there is "room for the averaging effects of Laplace's equation to occur".

Remark (what Harnack's inequality says).

1. *The constant does not depend on the function.* C is determined by the set V (and the fixed domain U): it is decided *before* any harmonic function is chosen. The same C works for *every* nonnegative harmonic function u in U.

2. *All values are comparable.* For any two points $x, y \in V$, $u(x)$ and $u(y)$ differ at most by the factor C. For example, if $u(y) = 1000$ at one point of V, then $u(x) \ge 1000/C$ at *every* point of V; and if $u(x) = 0.001$ at one point, then $u \le 0.001\, C$ everywhere in V.

3. *Why "nonnegative" is needed.* For $u(x) = x_1$ (harmonic, but changing sign) we have $\inf_V u < 0 < \sup_V u$ if V contains points on both sides of $\{x_1 = 0\}$, and no inequality $\sup u \le C \inf u$ with $C > 0$ can hold.

4. *Why $V \subset\subset U$ is needed.* The averaging argument needs balls around points of V which stay inside U. If V were allowed to approach ∂U, the balls would shrink and the constant would blow up.

5. *Why "connected" is needed.* If V had two separate pieces, information could not travel from one piece to the other through V; the proof passes from point to point through a chain of overlapping balls inside V.

Lemma 2.3 (Chain of balls)**.** *Let $V \subset \mathbb{R}^n$ be open and connected with $\bar{V}$ compact, and let $\rho > 0$. Then there exist finitely many open balls $B_1, \dots, B_N$ of radius ρ, centered at points of $\bar{V}$ and covering $\bar{V}$, with the following property: for any $x, y \in V$ there are distinct indices $i_0, i_1, \dots, i_m$ (so $m+1 \le N$) such that*

$$x \in B_{i_0}, \qquad y \in B_{i_m}, \qquad B_{i_{k-1}} \cap B_{i_k} \cap V \neq \emptyset \quad (k = 1, \dots, m).$$

Proof. Finite cover. The open balls $B^\circ(z,\rho)$, $z \in \bar{V}$, cover $\bar{V}$. Since $\bar{V}$ is compact, finitely many of them, $B_1, \dots, B_N$, already cover $\bar{V}$. Each B_j meets V: its center lies in $\bar{V}$, so it is a limit of points of V, and some of these are within distance ρ of the center.

Linked balls. Call two indices j, l *linked* if there is a chain $j = j_0, j_1, \dots, j_p = l$ with $B_{j_{q-1}} \cap B_{j_q} \cap V \neq \emptyset$ for each q. This is an equivalence relation (chains can be reversed and joined). Let $\mathcal{L}$ be the set of indices linked to 1, and put

$$A = \bigcup_{j \in \mathcal{L}} (B_j \cap V), \qquad A' = \bigcup_{j \notin \mathcal{L}} (B_j \cap V).$$

Both are open (unions of open sets), $A \cup A' = V$ (the balls cover V), and $A \ni$ a point of $B_1 \cap V$, so $A \neq \emptyset$.

A and A' are disjoint. If $z \in B_j \cap B_l \cap V$ with $j \in \mathcal{L}$ and $l \notin \mathcal{L}$, then j and l are directly linked (the one-step chain j, l), so $l \in \mathcal{L}$, a contradiction.

Conclusion. V is connected, $V = A \cup A'$ with A, A' disjoint and open and $A \neq \emptyset$; hence $A' = \emptyset$. If some index l were not in $\mathcal{L}$, then $\emptyset \neq B_l \cap V \subset A'$; so all indices are linked to 1, hence to each other. Given $x, y \in V$, pick i_0 with $x \in B_{i_0}$ and i' with $y \in B_{i'}$, and a chain from i_0 to i'. If an index occurs twice in the chain, delete the part of the chain between the two occurrences; repeating this, we obtain a chain with distinct indices. □

Proof of Theorem 11. Let $u \geq 0$ be harmonic in U, and let $V \subset\subset U$ be open and connected.

Step 0: choice of r. $\bar{V}$ is compact, ∂U is closed (and nonempty since U is bounded), and they are disjoint because $\bar{V} \subset U$. A compact set and a disjoint closed set have positive distance, so $\operatorname{dist}(V, \partial U) > 0$. Let

$$r := \tfrac{1}{4} \operatorname{dist}(V, \partial U).$$

Then every $x \in V$ has $\operatorname{dist}(x, \partial U) \geq 4r > 2r$, so $B(x, 2r) \subset U$ (Lemma 0.2). *This is the purpose of the factor $\frac{1}{4}$*: for every point of V we have room for a ball of radius $2r$ inside U (with some room to spare).

Step 1: comparing $u(x)$ and $u(y)$ for nearby points. Choose $x, y \in V$, $|x - y| \leq r$.

- $B(x, 2r) \subset U$ by Step 0, so the mean-value formula holds on it.
- $B(y, r) \subset B(x, 2r)$: if $|z - y| \leq r$, then $|z - x| \leq |z - y| + |y - x| \leq r + r = 2r$.
- Since $u \geq 0$, integrating over the smaller set $B(y, r)$ gives a smaller value: $\int_{B(x,2r)} u \, dz \geq \int_{B(y,r)} u \, dz$. (We drop the integral over $B(x, 2r) \setminus B(y, r)$, which is ≥ 0. *Here the nonnegativity of u is used.*)

Therefore

$$\begin{aligned} u(x) &= \fint_{B(x,2r)} u \, dz = \frac{1}{\alpha(n) 2^n r^n} \int_{B(x,2r)} u \, dz \geq \frac{1}{\alpha(n) 2^n r^n} \int_{B(y,r)} u \, dz \\ &= \frac{1}{2^n} \fint_{B(y,r)} u \, dz = \frac{1}{2^n} u(y). \end{aligned}$$

In the last step we used the mean-value formula on $B(y, r)$, which lies in U (it lies in $B(x, 2r)$).

Step 2: interchanging the roles of x and y. The point y is also in V, so by Step 0, $B(y, 2r) \subset U$ as well, and $B(x, r) \subset B(y, 2r)$ (same triangle inequality). Thus the argument of Step 1 works with x and y interchanged, giving $u(y) \geq 2^{-n} u(x)$. Altogether,

$$2^n u(y) \geq u(x) \geq \frac{1}{2^n} u(y) \qquad \text{if } x, y \in V, \ |x - y| \leq r.$$

Step 3: chains of balls. Since V is connected and $\bar{V}$ is compact, Lemma 2.3 with $\rho = r/2$ gives finitely many open balls $B_1, \dots, B_N$ of radius $r/2$ covering $\bar{V}$, such that any two points of V are joined by a chain of these balls with consecutive balls meeting inside V. The number N depends only on V (and r, which depends only on V and U), *not on* u.

Now let $x, y \in V$. Take the chain $B_{i_0}, \dots, B_{i_m}$ from Lemma 2.3, and choose points

$$p_0 = x, \qquad p_k \in B_{i_{k-1}} \cap B_{i_k} \cap V \ (k = 1, \dots, m), \qquad p_{m+1} = y.$$

All p_k lie in V. For each $k = 0, \dots, m$, the two consecutive points p_k and p_{k+1} both lie in the ball B_{i_k} (check: $p_0 = x \in B_{i_0}$; $p_k \in B_{i_k}$ for $k \geq 1$; $p_{k+1} \in B_{i_k}$ for $k < m$; $p_{m+1} = y \in B_{i_m}$). A ball of radius $r/2$ has diameter r, so $|p_k - p_{k+1}| < r$. By Step 2, $u(p_k) \geq 2^{-n} u(p_{k+1})$ for $k = 0, \dots, m$. Chaining these $m + 1$ inequalities (all quantities are ≥ 0),

$$u(x) = u(p_0) \geq \frac{1}{2^n} u(p_1) \geq \frac{1}{2^{2n}} u(p_2) \geq \cdots \geq \frac{1}{2^{n(m+1)}} u(p_{m+1}) = \frac{1}{2^{n(m+1)}} u(y) \geq \frac{1}{2^{nN}} u(y),$$

since $m + 1 \leq N$.

Step 4: conclusion. Let $C := 2^{nN}$, which depends only on V. We have shown $u(y) \leq C u(x)$ for all $x, y \in V$; by symmetry also $u(x) \leq C u(y)$, i.e. $\frac{1}{C} u(y) \leq u(x) \leq C u(y)$. Fix x and take the supremum over $y \in V$: $\sup_V u \leq C u(x)$. Now take the infimum over $x \in V$: $\sup_V u \leq C \inf_V u$. $\square$